3

REVISITING MILGRAM'S 1978 "RESPONSE TO INTRUSION INTO WAITING LINES" EXPERIMENT

This analysis shows that cost alone cannot account for all our results; there is an underlying structure to the situation, linked to the linear spatial configuration of the queue.

Milgram et al. (1986, p. 688), with permission from the American Psychological Association

Dr. Stanley Milgram was prodigious at creating or facilitating the design and implementation of what are nothing less than brilliantly devised experiments. One of those experiments, conducted by his graduate students in the spring of 1978 and published shortly after his death (Milgram et al., 1986), relates to how queue members in typical New York locations react to a simultaneous intrusion by one or two student confederate members. Experimental trials were performed 20 or more times for each of six distinct conditions (zero, one, or two non-reacting buffers, and one or two intruders) using 17 locations in New York City where queues regularly formed. Data contained in Milgram et al. (1986) along with summary data and three of the original 1978 graduate student reports from the Stanley Milgram Papers (MS 1406) allow further analysis and model development to continue from the previous chapter. Results of this effort, using exponential encoding, lead to a significantly expanded and coherent interpretation of social interaction and structure within the linear spatial configuration of the queue as a social system.

3.1 Establishing an Analytic Model of Intragroup Social Impact

Six years after the article by Milgram et al. (1986) is published, Schmitt et al. (1992) publish an equally important paper further exploring the queue as a social system. The importance of this paper is that it provides a critical complement to the 1986 paper by Stanley Milgram et al. The paper by Schmitt et al. involves four studies − three were attitudinal questionnaires, and the fourth was a field experiment conducted at New York's Grand Central Station. It is the fourth study – the field experiment – that will be made use of.

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In the fourth study, Schmitt et al. (1992) collect data consisting of 120 observations (four conditions with 30 observations each). Schmitt and his team use three control factors in this field experiment to focus specifically on the individual queue member who is intruded in front of by a student confederate. The control factors, designed to minimize influence from queue members near the member of interest, are:

1. All observations are from a single social situation – Grand Central railroad station ticket counter, New York City.
2. There is always a confederate in queue ahead of the intrusion point to eliminate reaction from members ahead of the intruder.
3. In the high social obligation case, two people in queue behind the member of interest are confederates who join the queue right after the member of interest joins.

In addition to the three control factors, Schmitt et al. (1992) consider four intrusion scenarios (i.e., conditions). The four scenarios in their field experiment are:

1. A confederate intruding in front of the last queue member in queue (illegitimate/low social obligation).
2. A confederate intruding in front of the third from the last member in queue (illegitimate/high social obligation).
3. A confederate joining the queue in front of the last queue member in queue, while acting as a friend to the student confederate who is in line ahead of the queue member (legitimate/low social obligation).
4. A confederate joining the queue in front of the third-to-last queue member in queue, while acting as a friend to the student confederate who is in line ahead of the queue member (legitimate/high social obligation).

Reaction by the queue member of interest, directed at the intruder, falls into one of three reaction categories directed at the intruder. The reaction categories, used also by Milgram et al. (1986), are indirect objection (i.e., body language or indirect comments), direct verbal objection, or physical action used to gain the attention of and eject the intruder. Probability of reaction for each condition is shown in Table 3.1.

Findings from this and the Milgram intrusion data to be presented is that the greater the sensation magnitude representing the group member of interest's dissonance, the

TABLE 3.1 Probability of a Reaction, Given the Experimental Condition (Schmitt et al., 1992, p. 813 Table 1) with Permission from the American Psychological Association

Experimental Condition	*P(Reaction) =*
Illegitimate Intruder	
Low Social Obligation	0.366
High Social Obligation	0.600
Legitimate Intruder (Acts as Friend)	
Low Social Obligation	0.100
High Social Obligation	0.133

greater the probability of direct verbal and physical reaction by that group member. This is an area of study that could go further in clarifying the severity of an individual's response as a function of increased dissonance. To continue, beginning with the two illegitimate experimental conditions, the probability of the queue member of interest reacting when no one is behind him or her is 0.366, compared to 0.6 when there are two people behind the member of interest. We have observed that Fechner's law provides an excellent representation of how, in the crowd-gathering experiment, the number of confederates looking up influences the reaction probability of an observer. It has also been argued (Dehaene et al., 2003, 2008) that the human brain perceives nonsymbolic numbers on a logarithmic scale, so it may be conjectured that some variation to the generalization of Fechner's Law given in Equation 2.2 will apply to the number of queue members behind the member of interest. To use Fechner's Law under exponential encoding as defined in Equation 2.3, and to account for queue members behind the member of interest, an additional concept from cognitive dissonance theory (Festinger, 1957, pp. 183–185) must now be introduced – that of group attraction.

> *The data showed, again, that the greater the [group member] attraction to the group, the greater was the degree to which the members tried to influence one another. In other words, the greater the magnitude of dissonance, the stronger the attempt to reduce this dissonance by changing the opinion of the person who disagreed.*
>
> *Leon Festinger (1957, p. 185, with permission from Stanford University Press)*

If group attraction is based on strength of belief in the group's social norms, and the person who disagreed now becomes the intruder who is now part of the group, then any uncertainty regarding group attraction must become part of the social queue equation. Letting group attraction be synonymous with group cohesiveness, the more cohesive the group, the greater the pressure on its members to react to dissonant-causing deviations of group social norms. In ad hoc groups formed for a specific purpose, members not familiar with one another may be less certain regarding intra-group attraction. A group-based noise random variable N_{UA} is used to account for this uncertainty of attraction. Let $p, m \in \mathbb{Z}^0$ where m represents the intensity, or number of people behind the queue member of interest, and p the number of intruders. Building on Equation 2.3, there exists a function $x(m)$ in the format of Equation 1.4 such that

$$0 = G_{ES}\left[\overline{es}_o, x(m)\right] \le G_{ES}\left[\overline{es}_p, x(0)\right]$$
$$= G_{ES}\left(\overline{es}_p\right) \le G_{ES}\left[\overline{es}_p, x(m)\right] \le G_{ES}\left[\overline{es}_p, x(\infty)\right] = 1, or$$

$$G_{ES}(\overline{es}_{p,m}) = 1 - e^{-\overline{es}_p \cdot x(m)} = 1 - e^{-\overline{es}_p \cdot ln\left(e \cdot \frac{m+N_{UA}}{N_{UA}}\right)}. \qquad \textit{Equation } 3.1$$

Now, instead of number of confederates looking up as in the crowd-gathering experiment, we are interested in number of confederates intruding and the number of members behind the queue member of interest who the intruder cuts in front of. With that distinction, let us try an approach similar to what was done in Chapter 2 using Table 3.1 data.

For illegitimate intrusions, the geometric mean for uncertainty of attraction $\widehat{N}_{UA}$ may be derived using the low and high social obligation cases in Table 3.1 by inverting Equation 3.1, the CDF $G_{ES}(\overline{es}_{P,M})$, such that

$$\frac{Illegitimate\,High\,social\,obligation}{Illegitimate\,Low\,social\,obligation} = \frac{-ln(1-.6)}{-ln(1-.366)} = \frac{0.916}{0.456} =$$

$$2.009 = \frac{\overline{es}\cdot ln\left(\dfrac{1+\widehat{N}_0}{\widehat{N}_0}\right)\cdot ln\left(e\cdot\dfrac{2+\widehat{N}_{UA}}{\widehat{N}_{UA}}\right)}{\overline{es}\cdot ln\left(\dfrac{1+\widehat{N}_0}{\widehat{N}_0}\right)\cdot ln\left(e\cdot\dfrac{0+\widehat{N}_{UA}}{\widehat{N}_{UA}}\right)} = ln\left(e\cdot\frac{2+\widehat{N}_{UA}}{\widehat{N}_{UA}}\right).$$

Solving for $\widehat{N}_{UA}$ results in $\widehat{N}_{UA} = 1.147$. The sensation magnitude sample mean $\overline{es}_p$ for this given social situation and set of social norms may be obtained using the illegitimate low social obligation condition where,

$$\begin{aligned} illegitimate\,(il)\,case\,sensation\,magnitude\,\overline{es}_{il,1,0} &= -ln\,(1-0.366) \\ &= 0.456. \end{aligned}$$

For the two illegitimate (il) cases, using Equation 3.1, we now have,

$$G_{ES}(\overline{es}_{il,1,0}) = 1-e^{-0.456\cdot ln(e)} = 0.366, and$$

$$G_{ES}(\overline{es}_{il,1,2}) = 1-e^{-0.456\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)} = 0.6.$$

which are the original values. Since the illegitimate and legitimate cases are for the same social situation and social norms, we will assume for now until more data comes along that group noise caused by uncertainty of attraction in the legitimate case is the same. Now, using the value $\widehat{N}_{UA} = 1.147$, we can determine an average for legitimate high and low social obligation sensation magnitude using Table 3.1 data to obtain a more accurate sample mean for legitimate (le) intrusion sensation magnitude $\overline{es}_{le,1,0}$,

$$(le)\,low\,social\,obligation\,\overline{es}_{le,1,0} = -ln(1-0.1) = 0.105.$$

Converting (le) high to (le) low social obligation;

$$\overline{es}_{le,1,0} = \frac{-ln(1-0.133)}{ln\left(e\cdot\dfrac{2+1.147}{1.147}\right)} = 0.071.$$

For the legitimate case, we then have the average,

$$(le)\ low\ social\ obligation\ \overline{es}_{le,1,0} = \frac{0.105 + 0.071}{2} = 0.088.$$

Therefore, for the two legitimate cases

$$G_{ES}(\overline{es}_{le,1,0}) = 1 - e^{-0.088 \cdot ln(e)}$$
$$= 0.084, which\ is\ within\ the\ 95\%\ credibility\ interval$$
$$(.036, .258).$$

$$G_{ES}(\overline{es}_{le,1,2}) = 1 - e^{-0.088 \cdot ln\left(e \cdot \frac{2+1.147}{1.147}\right)}$$
$$= 0.162, which\ is\ within\ the\ 95\%\ credibility\ interval$$
$$(.054, .298).$$

Given only a single intruder, there is currently insufficient information to calculate unit stimulus sensation magnitude $\overline{es}$ and social noise $\widehat{N}_0$ for either the illegitimate or legitimate case.

With Equation 3.1, representing amplification of social pressure on the queue member of interest to react as a function of members in queue behind him or her, and with $\overline{es}_{il,1,0}$ and $\widehat{N}_{UA}$ derived using Grand Central railroad station data, we are ready to analyze the more complex Milgram et al. (1986) intrusion data, consisting of 17 locational social situations, the use of one or two intruders, various numbers of queue members behind the member of interest, and having from zero to two non-reacting confederate buffers between the intruder and queue member of interest.

3.2 Milgram's "Response to Intrusion Into Waiting Lines": Reaction Summary and Associated Social Situations

Before analyzing Milgram's intrusion experiment, its structure and the social situations from which the data was taken must be clearly understood. Three graduate student reports graded by Dr. Milgram in 1978 were found at Stanley Milgram Papers (MS 1406), Manuscripts and Archives, Yale University Library, along with supplementary raw summary data. The student reports and supplementary data clarify how the experiment was conducted and what was learned. In a few instances, data was found that is otherwise not available in the published article by Milgram et al. (1986). Sadly, few members of the larger experiment team are still alive, and those that are do not have the 126 coding data observation sheets (see example in Appendix A) which would allow for a more straightforward analysis of the overall set of experiments. These data sheets had information on every trial, including location of each observation, date, time, how each queue member reacted as a function of queue position, and how many queue members were in each observed queue. As a result, to address the latter, queueing theory is applied (or abused for the pure at heart) in Appendix A to approximate the actual mean length of the Condition 4 queue. The Condition 4 queue,

in 17 of its 23 trials, clearly extends beyond what is available in the raw summary data, which ends at the fifth queue member position behind the point of intrusion. The law of total probability and the chain rule from Bayesian statistics are heavily employed to approximate group statistics that cannot otherwise be evaluated directly due to the absence of coding data. All of this is to say that, as we wade into the math, the approach requires some assumptions which are implemented in a transparent and methodical manner. It is up to the reader to evaluate their merit in trying to do the best with what data exists. To allay any fears, this chapter will not finish with the statement, "more data is needed." It will end with a complete model prototype allowing quantitative evaluation against additional social experiments with a queue-like structure for either disproving or further validating the model as derived. This hopefully begins to address what Thomas Pettigrew was looking for in his reassessment commentary of social psychological theory (Pettigrew, 1986, pp. 169–195).

The upcoming calculations focus predominantly on data contained in both Milgram et al. (1986, p. 685, Table 3.2) and the Stanley Milgram Papers (MS 1406), Manuscripts and Archives, Yale University Library, student experiment reports of Raymond

TABLE 3.2 Original 1978 Raw Summary Reaction Data by Condition and Position. Note that Milgram et al. (1986) Data Takes Precedence When Any Differences are Present

Milgram et al (1986) Data	Breakdown of queue member actions as a function of queue condition and queue member position Q(P|N|M). P is the number of simulataneous intruders, N is the number of queue members or confederate non-reactive buffers between member of interest and intruder(s), and M is the number of queue members behind the member of interest.	Condition 1: (+1) - Q(1|0|*)	Condition 1: (+2) - Q(1|1|*)	Condition 1: (+3) - Q(1|2|*)	Condition 2: (+2) - Q(1|1|*) 1 Buffer	Condition 2: (+3) - Q(1|2|*) 1 Buffer	Condition 3: (+3) - Q(1|2|*) 2 Buffers	Condition 4: (+1) - Q(2|0|*)	Condition 4: (+2) - Q(2|1|*)	Condition 4: (+3) - Q(2|2|*)	Condition 5: (+2) - Q(2|1|*) 1 Buffer	Condition 5: (+3) - Q(2|2|*) 1 Buffer	Condition 6: (+3) - Q(2|2|*) 2 Buffers	Condition 6: (+4) - Q(2|3|*) 2 Buffers
	Probability of Member Reaction to Intruder Based on Experimental Data:	0.364	0.143	0.000	0.167	0.000	0.000	0.870	0.435	0.091	0.200	0.000	0.150	0.118
	# of Reactions Divided by Total Observations per Condition and Queue Member of Interest (+[N+1])	8/22=	2/14=	0/9 =	4/24=	0/15 =	0/20 =	20/23=	10/23=	2/22=	4/20=	0/15 =	3/20=	2/17=
Stanley Milgram Papers - Archives (MS 1406, Box 100, Folder 131), Yale University Library	# Physical Ejection of Intruder (PE)	1						6	2				1	1
	# Verbal Ejection of Intruder (VE)	4	1		1			9	7	3				1
	# Verbal Disapproval to Intruder (VDI)	3						3	1		1			
	# Nonverbal Dissaproval to Intruder (NDI)	↑	1		3			2		1	4		2	↑
		Reaction Counted Only When Directed at Intruder												
	# Verbal Dissaproval to Others (VDO)	2					1		2		1	1	2	1
	# Nonverbal Disapproval to Others (NDO)		2	1			1			1	2		1	
	# Undirected Dissaproval (UD)	2	0		4	1	1				3	2	4	1
	# No Reaction Detected (NRD)	7	10	8	15	14	17	2	10	17	9	12	10	12
	#Verbal/Nonverbal Approval (VNA)	3			1			1						

Unpublished data obtained from The Stanley Milgram Papers (MS 1406) Manuscripts and Archives, Yale University Library with permission from the Estate of Alexandra Milgram. Where Stanley Milgram Paper raw data differs slightly, precedence given to final data in Milgram et al (1986).

Toledo, David Nemiroff, and Christina Taylor who were graduate students under Stanley Milgram in 1978. Also used from the Stanley Milgram Papers (MS 1406) are four pages of rough notes written in preparation for the line intrusion experiment by then graduate student Joyce Wackenhut, raw summary queue member line position reaction data for each of the six conditions (individual trial data not available), and summary data regarding queue locations for each of the six conditions. After presenting pertinent summary data from the experiment as a function of experiment condition, the order of analysis taken for each of the six conditions is performed to support derivation of necessary values and additional analytic tools. Analysis begins with Condition 1, composed of 22 trials from a total of eight different queue locations. In Condition 1, a single confederate intruder is observed entering a line at a position (termed queue position zero) having typically five or less queue members after the intrusion point. Each queue member position after the intrusion point is labeled as position (+1), (+2), (+3), (+4), and (+5). Queue members forward of the intrusion point are labeled as positions (−1), (−2), (−3), and (−4) and are not analyzed given the additional variables which cannot be accounted for such as did they even notice. Both Milgram et al. (1986) and Schmitt et al. (1992) provide more detailed discussion on the latter but neither reach a definitive conclusion.

Table 3.2 is consolidated from the original 1978 raw summary data obtained from Stanley Milgram Papers (MS 1406). Any reaction by the queue member of interest directed at the intruder is counted as a reaction. For example, "Condition 1: $(+1) - Q(1 \mid 0 \mid *)$" indicates number of reactions from the Condition 1 set of observations having a single intruder "1," with no members "0" between the intruder and the member of interest who is therefore at the (+1) position, and with various numbers "*" of queue members behind the member of interest. In this case, of the 22 observations, 8 of the 22 queue members at the (+1) position react directly to the intruder. As another example, Condition 4:$(+3) - Q(2 \mid 2 \mid *)$ indicates probability of reaction resulting from two simultaneous intruders "2," by queue member position (+3) (i.e., two intermediate queue members fill the (+1) and (+2) positions in front of the position of interest), with various numbers "*" of queue members behind the position of interest. Of the 22 trials where two intruders simultaneously intrude at position (0), the (+3) position queue members with two queue members between themselves and the two intruders are observed to react twice out of 22 observations. Looking at Table 3.2 for Condition 4: (+1), it is readily apparent there are more reactions in this condition than for Condition 1: (+1). As importantly, the severity of reactions in Condition 1: (+1) is mostly verbal disapproval or verbal ejection. In Condition 4: (+1), the severity of member reaction moves up predominantly to verbal and physical ejection of the intruders. This could be due to the number of queue members behind the queue member of interest, it could be due to the number of intruders, or it could be due to the social situation (i.e., importance or limited availability of the queue resource commodity). As we will discover shortly, it is due to all three. Basically, as sensation magnitude increases, so does member aggression. This supports Axiom 1 and the quote by Festinger (1957, p. 18, with permission from Stanford University Press), "*The strength of the pressures to reduce the dissonance is a function of the magnitude of the dissonance*." Something similar is obviously occurring in the queue as in the crowd data experiment. As seen in the crowd-gathering data, the larger the crowd that deviates from a social norm, the

greater the dissonance felt by the observer and the greater the pressure on average to look up and reduce the dissonance.

Indication has been provided that the queue's social situation plays a role in the queue member's sensation magnitude resulting from an intrusion. This supports the person−situation interaction equation (Lewin, 1997, p. 187; Burnes and Cooke, 2013, pp. 412–413). To lay the foundation for this claim, locations and situations of queues are listed in Table 3.3 by Condition. Focus should be placed on the social situation, not the location. For instance, we may ask if waiting in line for a train ticket, having a relatively strict time schedule, is the same as waiting in line for a snow cone on our way

TABLE 3.3 Queue Experiment Locations and Frequency of Use by Condition

Locations that Queue Intrusion Data was Observed From:	Milgram et al (1986) Condition 1	Milgram et al (1986) Condition 2	Milgram et al (1986) Condition 3	Milgram et al (1986) Condition 4	Milgram et al (1986) Condition 5	Milgram et al (1986) Condition 6	Schmitt et al (1992) - Illegitimate	Schmitt et al (1992) - Legitimate
Bank (86)	2	0	2	0	1	0	0	0
Choc Full of Nuts (42)	0	0	0	0	1	0	0	0
Gyros Restaurant (42)	0	0	1	0	0	0	0	0
Health Food Store (GC)	0	0	1	0	0	0	0	0
Information (GC)	1	1	0	2	1	0	0	0
Investment (GC)	2	0	2	1	0	0	0	0
Passport (50)	1	0	1	0	0	0	0	0
Post Office (43)	0	0	0	0	4	0	0	0
Snow Cone Store (GC)	0	1	0	0	0	0	0	0
Zaro's Restaurant (34)	0	0	1	0	0	0	0	0
Token Booth (GC)	0	3	3	0	0	0	0	0
Ticketron (GC)	4	0	2	1	1	0	0	0
	Low Unit Stimulus Sensation Magnitude Categories (es_L) - White							
Bus Tickets (PA)	0	0	0	0	0	7	0	0
Bus Boarding (PA)	0	0	0	0	0	8	0	0
Airport (L)	1	0	2	0	0	0	0	0
Ticket Rail (GC)	8	16	3	18	6	4	30	30
Off-track Betting (OTB) (GC)	3	3	2	1	6	1	0	0
	High Unit Stimulus Sensation Magnitude Categories (es_H) - Gray Shaded							
Total Number of Observations:	22	24	20	23	20	20	30	30
A - # Hypothesized High Unit Stimulus Sensation Magnitude	12	19	7	19	12	20	30	0
B - # Hypothesized Low Unit Stimulus Sensation Magnitude	10	5	13	4	8	0	0	30
For Weighting:								
Pr(A) =	0.545	0.792	0.350	0.826	0.600	1.000	1.000	0.000
Pr(B) =	0.455	0.208	0.650	0.174	0.400	0.000	0.000	1.000

(L) - LaGuardia, TWA Boarding Line (PA) - Port Authority (GC) - Grand Central Terminal

(#) - Street Locations in Manhatten City, NY (OTB) - Off-track betting parlor

Note * : Graduate student David Nemiroff indicates for Condition 4, that two of the three non-reactions were at the GC Information Booth and that the Information Booth queue was amorphous (i.e. not well structured as a queue).

Data obtained from the Stanley Milgram Papers (MS 1406) Manuscripts and Archives, Yale University Library with permission from the Estate of Alexandra Milgram.

home. It is hypothesized here that limited resources (i.e., time to get the desired train before it departs, selection of a "good" seat when boarding a bus, time available to make a bet on your favorite horse before the race starts) are of greater importance and hence result in greater extent social deviation caused by an intruder than compared to queues with abundant or relatively unlimited resources (i.e., time, snow cones, stamps at the post office, passport ID, etc.). Those queues proposed as higher unit stimulus sensation magnitude queues include the five gray shaded locations of Table 3.3 which include the bus terminal, airport, ticket rail, and off-track betting locations. On the other hand, the 12 queue locations above the gray shaded area in Table 3.3 support queues waiting for relatively unlimited resources in a more relaxed setting. For example, Condition 1 encompasses eight different locations, where 12 observations are in potentially high and 10 are in potentially low-unit stimulus sensation magnitude situations. Alternatively, 18 of 23 queue observations for Condition 4 are in the potentially high-unit stimulus sensation magnitude ticket rail queue at Grand Central Terminal, the same location as the data from Schmitt et al. (1992). With this background, we are now ready to begin quantitative analysis of the data.

3.3 Milgram's "Response to Intrusion Into Waiting Lines": Analysis and Model Development

We begin here with the tools developed and values derived from Schmitt et al. (1992), which allow us to begin quantitative evaluation of the Milgram et al. (1986) experiment. Using results derived from Schmitt et al. (1992) for $\overline{es}_{il,1}$ and $\widehat{N}_{UA}$, the single intruder sensation magnitude value for low sensation magnitude queues is derived using Condition 1: $(+1) - Q(1 \mid 0 \mid *)$. This in turn should support derivation of the two-intruder high- and low-unit sensation magnitude sample mean using Condition 4: $(+1) - Q(2 \mid 0 \mid *)$. If Corollary 2.3 is met, the results from these two conditions and associated situations may then be applied to the remaining queue conditions to improve sample mean values and validate model predictions regarding impact of queue members between the intruder(s) and position of interest. Once complete, a spreadsheet incorporating the final model and all conditions simultaneously is used to fine tune results using the mean least squares method. With that complete, comparison between theoretical predictions versus empirical results is made against the 95 percent credibility intervals based on empirical results.

3.3.1 Condition 1: (+1)-Q(1|0|M) Probability of Reaction Calculation

Table 3.4 represents data from Condition 1 of Milgram et al. (1986). The associated analysis focuses on this data. For now, define N as the number of queue members between the intruder(s) and the position of interest. The number of queue members $m \in M$ behind the position of interest is found in the macro variable $Q(P \mid N \mid M)$ of Table 3.4. This variable also identifies the associated fraction of trials in which this configuration exists as a function of queue member position. To understand how the data was obtained, consider $Q(1 \mid 0 \mid 0)$ at position $(+1)$. There were 22 trials observed, yet in position $(+2)$ there were only 14 trials having members in that position. That

TABLE 3.4 Condition 1 Empirical Data

$Q(P\|N\|M) = Q(P$ *intruders* $\|\ N$ *queue members between intruder and position of interest* $\|\ M$ *queue members behind queue position of interest*$) =$ *Position Probability of Occurrence*				
Position (-1)	Position (-2)	Position (-3)	Position (-4)	Front of Queue
5/22	**1/22**	**0/21**	**0/5**	# React /# Opportunities
1 Intruder (Condition 1)	**Notes**: 1) Appendix A has k=1 and l_{100}=64.9: theoretical L = 1.8 ≈ documented M_{mean} = 1.7 in the Q(1|0|*) configuration: No extrapolation for "M" necessary.			
Position (+1)	Position (+2)	Position (+3)	Position (+4)	Position (+5)
8/22	**2/14**	**0/9**	**0/7**	**0/7**
$Q(1\|0\|4) = 7/22$	$Q(1\|1\|3) = 7/14$	$Q(1\|2\|2) = 7/9$	$Q(1\|3\|1) = 7/7$	$Q(1\|4\|0) = 7/7$
$Q(1\|0\|3) = 0/22$	$Q(1\|1\|2) = 0/14$	$Q(1\|2\|1) = 0/9$	$Q(1\|3\|0) = 0/7$	Milgram et al (1986)
$Q(1\|0\|2) = 2/22$	$Q(1\|1\|1) = 2/14$	$Q(1\|2\|0) = 2/9$	supplemented by unpublished data obtained from	
$Q(1\|0\|1) = 5/22$	$Q(1\|1\|0) = 5/14$	The Stanley Milgram Papers (MS 1406), Series No. IV, Box No. 100, Folder No. 131		
$Q(1\|0\|0) = 8/22$	Manuscripts and Archives, Yale University Library with copyright permission from the Estate of Alexandra Milgram.			

means that there were $22 - 14 = 8$ trials out of 22 with zero queue members behind the (+1) position of interest. For position (+3), there were 9 trials out of 22 where the (+3) position had a member in it. Since the (+2) position had 14 trial observations with a member in it, that means there were $14 - 9 = 5$ trials where there were no queue members behind the (+2) position (i.e., $Q(1\,|\,1\,|\,0) = 5/14$). The same process is used to obtain the remaining values.

Let us start with the proposed sample mean sensation magnitude value $\overline{es}_{il,1} = 0.456$ for illegitimate intrusions and uncertainty of attraction $\widehat{N}_{UA} = 1.147$ as was derived from the Schmitt et al. (1992) data. Since the illegitimate intrusion data was obtained at the Grand Central Terminal ticket counter, the result applies to high-unit stimulus sensation magnitude queues. Define $\overline{es}_{H,1,0} = \overline{es}_{H,1} = 0.456$ as the sensation magnitude value for high (H) unit sensation magnitude queues with one intruder, zero members between the intruder and position of interest, and zero members behind the position of interest. Similarly, define the low (L) queue unit sensation magnitude value as $\overline{es}_L$. Given $\overline{es}_{H,1} = 0.456$, $\widehat{N}_{UA} = 1.147$, with Condition 1: (+1) results, we are now able to derive $\overline{es}_{L,1,m}$ using:

$$\overline{es}_{L,1,m} = \overline{es}_{L,1} \cdot ln\left(e \cdot \frac{m + \widehat{N}_{UA}}{\widehat{N}_{UA}} \right)$$

Together, using the chain rule and the law of total probability for Condition 1: (+1), we obtain the probability of reaction at the (+1) queue position $P(R_{+1})$ for unit sensation magnitude $\overline{es}_I, I \in \{L, H\}$. The probability of $P(\overline{es}_I)$ is obtained from Table 3.3. Using the probability of queue member position $P(m) = Q(1\,|\,0\,|\,m)$ from Table 3.4 where $m \in M = \{0,1,2,3,4\}$,

$$\begin{aligned} P(R_{+1}) &= \sum_{I \in \{L,H\}} \sum_{m \in M} P\left(ES \le \overline{es}_{I,1,m}, \overline{es}_I, m\right) \\ &= \sum_{I \in \{L,H\}} \sum_{m \in M} G_{ES}\left(\overline{es}_{I,1,m}\right) \cdot P\left(\overline{es}_I \,|\, m\right) \cdot P\left(m\right) \end{aligned}$$

$$= \sum_{I\in\{L,H\}} \sum_{m\in M} \left[1 - e^{-\overline{es}_I ln\left(\frac{1+\bar{N}_0}{\bar{N}_0}\right)\cdot ln\left(e\cdot\frac{m+\bar{N}_{UA}}{\bar{N}_{UA}}\right)} \right] \cdot P\left(\overline{es}_I \mid m\right)\cdot P(m).$$

For our purposes $P(R_{+1})$, the probability of reaction at the $(+1)$ position, may be stated as

$$P\left(R_{+1}\right) = \sum_{I\in\{H,L\}} \sum_{m\in M} P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_I, Q\left(1|0|m\right)\right]$$
$$= \sum_{I\in\{H,L\}} \sum_{m\in M} G_{ES}\left(\overline{es}_{I,1,m}\right)\cdot P\left[\overline{es}_I \mid Q\left(1|0|m\right)\right] \cdot P\left[Q\left(1|0|m\right)\right]$$

The Condition $1{:}(+1)$ probability of reaction may now be calculated using Tables 3.3 and 3.4. Since no remaining experimental data exists to do otherwise, we will necessarily assume throughout this and subsequent analyses that the proportion of high and low unit stimulus sensation magnitude queues is independent of *m*, implying $P\left[\overline{es}_I \mid Q\left(1 \mid 0 \mid m\right)\right] = P\left[\overline{es}_I\right]$.

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_H, Q\left(1|0|0\right)\right] = \left(1 - e^{-0.456}\right)\cdot\left(\frac{12\ \overline{es}_H\ Trials}{22\ Trials}\right)\cdot\left(\frac{8}{22}\right)$$
$$= 0.073$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_H, Q\left(1|0|1\right)\right] = \left(1 - e^{-0.456\cdot ln\left(e\cdot\frac{1+1.147}{1.147}\right)}\right)\cdot\left(\frac{12}{22}\right)\cdot\left(\frac{5}{22}\right)$$
$$= 0.065$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_H, Q\left(1|0|2\right)\right] = \left(1 - e^{-0.456\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)}\right)\cdot\left(\frac{12}{22}\right)\cdot\left(\frac{2}{22}\right)$$
$$= 0.030$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_H, Q(1|0|3)\right] = \left(1 - e^{-0.456 \cdot ln\left(e \cdot \frac{3+1.147}{1.147}\right)}\right) \cdot \left(\frac{12}{22}\right) \cdot \left(\frac{0}{22}\right)$$
$$= 0.0$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_H, Q(1|0|4)\right] = \left(1 - e^{-0.456 \cdot ln\left(e \cdot \frac{4+1.147}{1.147}\right)}\right) \cdot \left(\frac{12}{22}\right) \cdot \left(\frac{7}{22}\right)$$
$$= 0.118$$

To match $P(R_{+1}) = \frac{8}{22} = 0.364$ from Condition 1: (+1), a spreadsheet calculation was done to find that $\overline{es}_{L,1,0} = 0.114$. Using $\overline{es}_{L,1,0} = 0.114$, we continue with

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_L, Q(1|0|0)\right] = \left(1 - e^{-0.114}\right) \cdot \left(\frac{10\,\overline{es}_L\ Trials}{22\ Trials}\right) \cdot \left(\frac{8}{22}\right)$$
$$= 0.018$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_L, Q(1|0|1)\right] = \left(1 - e^{-0.114 \cdot ln\left(e \cdot \frac{1+1.147}{1.147}\right)}\right) \cdot \left(\frac{10}{22}\right) \cdot \left(\frac{5}{22}\right)$$
$$= 0.017$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_L, Q(1|0|2)\right] = \left(1 - e^{-0.114 \cdot ln\left(e \cdot \frac{2+1.147}{1.147}\right)}\right) \cdot \left(\frac{10}{22}\right) \cdot \left(\frac{2}{22}\right)$$
$$= 0.008$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_L, Q(1|0|3)\right] = \left(1 - e^{-0.114 \cdot ln\left(e \cdot \frac{3+1.147}{1.147}\right)}\right) \cdot \left(\frac{10}{22}\right) \cdot \left(\frac{0}{22}\right)$$
$$= 0.0$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_L, Q(1|0|4)\right] = \left(1 - e^{-0.114 \cdot ln\left(e \cdot \frac{4+1.147}{1.147}\right)}\right) \cdot \left(\frac{10}{22}\right) \cdot \left(\frac{7}{22}\right)$$
$$= 0.036$$

From this, and accounting for round-off error,

$$P(R_{+1}) = \sum_{I\in\{H,L\}} \sum_{m\in M} P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_I, Q(1\,|\,0\,|\,m)\right]$$
$$= \sum_{I\in\{H,L\}} \sum_{m\in M} G_{ES}\left(\overline{es}_{I,1,m}\right)\cdot P\left(\overline{es}_I\right)\cdot P\left[Q(1\,|\,0\,|\,m)\right] = 0.364$$

Condition 1: (+1) has now been used to systematically determine a sample mean for the low sensation magnitude value $\overline{es}_{L,1,0} = 0.114$. That there are only two categories, low- and high-unit stimulus sensation magnitude queues, is probably overly simplistic, but we need to begin somewhere, and having a dividing line between limited resource (ticket and time) and perceptually unlimited resource is a good starting point.

3.3.2 Condition 4 – Probability of Reaction Calculation for Q(2|0|M)

To find the sensation magnitude value for a New York City queue with two intruders, we begin with Condition 4: (+1). From Schmitt et al. (1992), the evaluation of the sample-mean sensation magnitude for Grand Central Terminal using 60 trials was $\overline{es}_{H,1,0} = 0.456$ with $\widehat{N}_{UA} = 1.147$. From Section 3.3.1, using Condition 1: (+1) it was found that

$$\overline{es}_{L,1,0} = \overline{es}_L \cdot ln\left(\frac{1+\widehat{N}_0}{\widehat{N}_0}\right) = \frac{\overline{es}_H \cdot ln\left(\frac{1+\widehat{N}_0}{\widehat{N}_0}\right)}{4.0} \; so \; \overline{es}_L = \frac{\overline{es}_H}{4.0},$$

$$and \; in \; similar \; manner \; \overline{es}_{L,2,0} = \frac{\overline{es}_{H,2,0}}{4.0}.$$

With Condition 4: (+1) results provided in Table 3.5, there is enough information now to solve for $\overline{es}_{H,2,0}$ in a New York City high-unit stimulus sensation magnitude queue situation using data from Milgram et al. (1986). The approach taken, using the same process as for Condition 1: (+1), is simple. Since the low-unit stimulus sensation magnitude queue is one-quarter the value of the high unit stimulus sensation magnitude queue, all that needs to be done is to find the value for $\overline{es}_{H,2,0}$, using a spreadsheet, such that after simultaneous summation of the equations, the result $P(R_{+1}) = 0.870$ is reached. Once the value for $\overline{es}_{\mathrm{H},2,0}$ is derived, if Corollary 2.3 is met, then the geometric mean $\widehat{N}_0$ and sample mean value $\overline{es}\left(\widehat{N}_0\right)$ may be derived, given values for both $\overline{es}_{H,2,0}$ *and* $\overline{es}_{H,1,0}$ have been successfully determined.

Before proceeding, the "Notes" section of Table 3.5 should be reviewed along with Appendix A to understand the rationale and mathematics behind approximating the actual Condition 4: (+1) queue length in which 17 trials had $m \in M, m \ge 4$ per the summary data. As with Condition 1, let $m \in M = \{0,1,2,3,11.3\}$ represent the elements of the set M containing the various number of queue members behind the queue member of interest.

TABLE 3.5 Condition 4 Empirical Data

$Q(P\|N\|M) = Q(P$ *intruders* $\mid N$ *queue members between intruder and position of interest* $\mid$ M *queue members behind queue position of interest*$) =$ *Position Probability of Occurrence*				
Position (-1)	Position (-2)	Position (-3)	Position (-4)	Front of Queue
3/23	**1/23**	**0/22**	**0/10**	# React /# Opportunities
2 Intruders (Condition 4)	**Notes**: 1) Appendix A has k=10 and l_{100}=95.2: theoretical M = 11.3 >> documented M_{mean} = 3.4 in the Q(2|0|*) configuration: Changed Q(2|0|4) to Q(2|0|11.3). 2) Original 1978 raw summary data and two graduate student reports indicate 10/22 at (+2), 4/22 at (+3), and 0/17 at the (+4) positions - differing with Milgram et al (1986) (i.e. 10/23, 2/22, 0/20). Milgram et al (1986) takes precedence.			
Position (+1)	Position (+2)	Position (+3)	Position (+4)	Position (+5)
20/23	**10/23**	**2/22**	**0/20**	**0/17**
$Q(2\|0\|11.3) = 17/23$	$Q(2\|1\|10.3) = 17/23$	$Q(2\|2\|9.3) = 17/22$	$Q(2\|3\|8.3) = 17/20$	$Q(2\|4\|7.3) = 17/17$
$Q(2\|0\|3) = 3/23$	$Q(2\|1\|2) = 3/23$	$Q(2\|2\|1) = 3/22$	$Q(2\|3\|0) = 3/20$	Milgram et al (1986)
$Q(2\|0\|2) = 2/23$	$Q(2\|1\|1) = 2/23$	$Q(2\|2\|0) = 2/22$	supplemented by unpublished data obtained from	
$Q(2\|0\|1) = 1/23$	$Q(2\|1\|0) = 1/23$	The Stanley Milgram Papers (MS 1406), Series No. IV, Box No. 100, Folder No. 131		
$Q(2\|0\|0) = 0/23$	Manuscripts and Archives, Yale University Library with copyright permission from the Estate of Alexandra Milgram.			

By spreadsheet, based on the following equations, it is found that $\overline{es}_{H,2,0} = \overline{es}_{H,2} = 1.005$ *and* $\overline{es}_{L,2,0} = \overline{es}_{L,2} = \dfrac{\overline{es}_{H,2}}{4.0} = 0.251$. As a function of $\overline{es}_{H,2}$, the equations used in the spreadsheet supporting a simultaneous solution are

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_H, Q(2|0|0)\right] = \left(1 - e^{-1.005}\right) \cdot \left(\frac{19\ \overline{es}_H\ Trials}{23\ Trials}\right) \cdot \left(\frac{0}{23}\right) = 0.0$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_H, Q(2|0|1)\right] = \left(1 - e^{-1.005 \cdot ln\left(e \cdot \frac{1+1.147}{1.147}\right)}\right) \cdot \left(\frac{19}{23}\right) \cdot \left(\frac{1}{23}\right) = 0.029$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_H, Q(2|0|2)\right] = \left(1 - e^{-1.005 \cdot ln\left(e \cdot \frac{2+1.147}{1.147}\right)}\right) \cdot \left(\frac{19}{23}\right) \cdot \left(\frac{2}{23}\right) = 0.062$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_H, Q(2|0|3)\right] = \left(1 - e^{-1.005 \cdot ln\left(e \cdot \frac{3+1.147}{1.147}\right)}\right) \cdot \left(\frac{19}{23}\right) \cdot \left(\frac{3}{23}\right) = 0.097$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_H, Q(2|0|11.3) \right]$$
$$= \left(1 - e^{-1.005 \cdot ln\left(e \cdot \frac{11.3+1.147}{1.147} \right)} \right) \cdot \left(\frac{19}{23} \right) \cdot \left(\frac{17}{23} \right) = 0.590$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_L, Q(2|0|0) \right]$$
$$= \left(1 - e^{-0.251} \right) \cdot \left(\frac{4\ \overline{es}_L\ Trials}{23\ Trials} \right) \cdot \left(\frac{0}{23} \right) = 0.0$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_L, Q(2|0|1) \right]$$
$$= \left(1 - e^{-0.251 \cdot ln\left(e \cdot \frac{1+1.147}{1.147} \right)} \right) \cdot \left(\frac{4}{23} \right) \cdot \left(\frac{1}{23} \right) = 0.003$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_L, Q(2|0|2) \right]$$
$$= \left(1 - e^{-0.251 \cdot ln\left(e \cdot \frac{2+1.147}{1.147} \right)} \right) \cdot \left(\frac{4}{23} \right) \cdot \left(\frac{2}{23} \right) = 0.006$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_L, Q(2|0|3) \right]$$
$$= \left(1 - e^{-0.251 \cdot ln\left(e \cdot \frac{3+1.147}{1.147} \right)} \right) \cdot \left(\frac{4}{23} \right) \cdot \left(\frac{3}{23} \right) = 0.010$$

$$P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_L, Q(2|0|11.3) \right]$$
$$= \left(1 - e^{-0.251 \cdot ln\left(e \cdot \frac{11.3+1.147}{1.147} \right)} \right) \cdot \left(\frac{4}{23} \right) \cdot \left(\frac{17}{23} \right) = 0.074$$

Assuming homogeneity of high- and low-dissonance magnitude queues as before and accounting for round-off error in the aforementioned,

$$P(R_{+1}) = \sum_{I\in\{H,L\}}\sum_{m\in M} P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_I, Q(2|0|m)\right]$$
$$= \sum_{I\in\{H,L\}}\sum_{m\in M} G_{ES}\left(\overline{es}_{I,2,m}\right)\cdot P\left(\overline{es}_I\right)\cdot P\left[Q(2|0|m)\right] = 0.870$$

Unfortunately, $\frac{\overline{es}_{H,2}}{\overline{es}_{H,1}} = \frac{1.005}{0.456} = 2.2 > \frac{2}{1}$. Hence, as per Corollary 2.3, there is no feasible solution for $\widehat{N}_0$ and hence for $\overline{es}\left(\widehat{N}_0\right)$ using Condition 4: (+1) data. It is worth mentioning that had one less (+1) reaction occurred from those 23 observations, the resultant sensation magnitude would have been $\overline{es}_{H,2} = 0.81$, and Corollary 2.3 would have been met.

Experiments take time to collect data, and for this experiment, the stress placed on the graduate students as intruders was significant – in some cases leading to feelings of nausea right before the intrusion. More data for this Condition would have reduced the credibility interval and should have brought the sensation magnitude value back within the Corollary 2.3 required region allowing for a feasible solution of $\widehat{N}_0$. But we should be grateful for what we have, and Condition 4: (+2) data will be extremely useful later.

Returning to our current problem, to find a feasible solution for $\widehat{N}_0$ and hence for $\overline{es}\left(\widehat{N}_0\right)$, the only remaining option is to derive $\overline{es}_{H,2}$ from Conditions 5 and/or 6 and determine if either or both meet Corollary 2.3 requirements. To do so, we must first derive the impact of the nonreactive buffer for Condition 2 using $\overline{es}_{H,1} = 0.456$, which is supported by 60 observations from Schmitt et al. (1992) and 12 observations from the Condition 1: (+1) data.

3.3.3 Impact of Nonreacting (NR) Confederate Buffers: Deriving z(|UI|) Using Conditions 2: (+2), 5: (+3), and 6: (+3)

Condition 2: (+2), 5: (+2), and 6: (+3) queues are chosen for the following reasons:

1. They are consistent with the raw data from Stanley Milgram Papers (MS 1406) and Milgram et al. (1986).
2. No extrapolation is required to determine the number of members behind the position of interest as was needed in Condition 4.
3. All queue members between the intruder(s) and position of interest are nonreactive confederate student buffers.
4. Each of the three conditions has an empirical probability of reaction greater than zero.

So far, intermediate queue members between the intruder(s) and the position of interest have not been considered. For $p \in P$ and $m \in M$, let us now replace N by defining

UI as the set containing ordered queue position reaction R_i or non-reaction NR_i information for each intermediate queue member at queue position $(+i) = 1, 2, \ldots, |UI|$, where $N = |UI|$ is the cardinality of UI. Equation 3.1 currently considers the null set $UI = \{\ \} \equiv \varnothing \Rightarrow |UI| = 0$ in which case $P(R_{+|UI|+1})$ indicates the $(+1)$ position of interest probability of reaction or $P(R_{+1})$.

Now we will develop Equation 3.1 further to address $|UI| > 0$. By observation of Table 3.2, probability of reaction decreases with each succeeding member position of interest after the intrusion point, regardless of whether each intermediate queue member reacts or does not react. Focusing only on non-reacting NR_i intermediate queue members for the moment, for some function $z(|UI|)$, $p > 0$, and $m \geq 0$, we observe $G_{ES}\left[\overline{es}_{p,m}, z(|UI|)\right]$ decreases as $|UI|$ increases. Furthermore, if $|UI| = 0$, then $G_{ES}\left[\overline{es}_{p.m}, z(0)\right] = G_{ES}\left(\overline{es}_{p,m}\right)$. It will be assumed for an infinite number of intermediate queue members $|UI| \to \infty$ that probability of reaction in a queue at the position of interest tends to zero such that $G_{ES}\left[\overline{es}_{p,m}, z(|UI|)\right] = 0$ *as* $|UI| \to \infty$. Finally, if there are no intruders (i.e., $p = 0$), probability of reaction $G_{ES}\left[\overline{es}_{0,m}, z(|UI|)\right] = 0$. Then for any given nonnegative integers $p, |UI|$, *and* m, constraints on $G_{ES}\left[\overline{es}_{p,m}, z(|UI|)\right]$ are summarized as

$$
\begin{aligned}
0 &= G_{ES}\left[\overline{es}_{0,m}, z(|UI|)\right] \\
&= \lim_{i \to \infty} G_{ES}\left[\overline{es}_{p,m}, z\left(\left|\{NR_1, NR_2, \ldots, NR_i\}\right|\right)\right] \\
&\leq G_{ES}\left[\overline{es}_{p,m}, z(|UI|)\right] \leq G_{ES}\left[\overline{es}_{p,m}, z(0)\right] \\
&= G_{ES}\left(\overline{es}_{p,m}\right).
\end{aligned}
$$

In words, the subject of interest is obtaining information from each intermediate position group member between him or her and the intruder. Those intermediate members who react indicate to the subject of interest that deviation from a social norm likely occurred. Their reaction, as judged by the next further observing intermediate member in queue, will range from either being sufficient or insufficient, the latter possibly warranting some further level of reaction. Those intermediate members who do not react indicate to the subject of interest that the social deviation was acceptable to them, or at least not sufficient to warrant a reaction as far as they were concerned. Each reaction or non-reaction by an intermediate queue member may be viewed as an increment of dissonance-reducing information made use of by the subject of interest to reduce uncertainty prior to making the choice of whether or not to react. When uncertainty of choice surrounding an event exists, it is hypothesized that dissonance-reducing information directed at relieving the uncertainty will reduce the event-induced dissonance.

When social events occur that are not congruent with how we believe people should act, based on our social norms for the given situation, then we experience emotional discomfort and associated dissonance. If the social event is ambiguous within the

given social situation, then we experience uncertainty and must make a choice regarding what action or nonaction provides the best opportunity to reduce the associated dissonance. Information is sought to make the best choice, and as Festinger (1957, pp. 177–202) indicates, the social group offers the best means for dissonance reduction by providing the necessary support from other group members who share similar social norms. Since the information we seek as per Axiom 1 is to reduce dissonance for the given socially deviant event and social situation, those in our particular social group, sharing common social norms, would offer the most reliable and pertinent sources of information concerning how to interpret the event and whether sufficient action has been taken.

We are focused on the ambiguous socially deviant event which creates uncertainty in how to respond to the given situation (Clark and Word, 1972). Reaction is one way to gain information and reduce dissonance; and observation of others in the group is a lower risk approach (see Axiom 2, Chapter 1). As a result, when not in the (+1) position, dissonance-reducing information pertinent to the generated uncertainty is required and probably desired when choosing which action increases the likelihood of successful dissonance reduction.

Using the concepts and nomenclature from Klir (2006, p. 7), the amount of A Priori uncertainty associated with the deviant social event occurring within a group having established and accepted social norms is U_{Pr}. After uncertainty-based dissonance-reducing information associated with reducing U_{Pr} has been obtained by the subject of interest through observing the non-reactions NR_i of intermediate group members, some residual A Posteriori uncertainty will likely remain. Denote this remaining A Posteriori uncertainty as U_{Po}. Define the uncertainty-based dissonance-reducing information set as $UI = \left\{\varnothing_0, NR_1, NR_2, \ldots, NR_{|UI|}\right\}$ in which $UI_i \in UI$ such that $UI_0 = \varnothing, UI_1 = NR_1$, and so on. Then $\bigcup_{i=0}^{|UI|} UI_i$ as the union of uncertainty-based information obtained through observing $|UI|$ intermediate member non-reactions, keeping in mind that $|\varnothing| = 0$. The total amount of uncertainty-based information gained as a function of uncertainty reduction is given by,

$$\begin{aligned}\left(A\ Priori\ uncertainty\right) - \left(A\ Posteriori\ uncertainty\right) &= U_{Pr} - U_{Po} \\ &= \bigcup_{i=0}^{|UI|} UI_i.\end{aligned}$$

This is equivalent to (Smith et al., 2015, pp. 99(d) & 106(e)),

$$U_{Po} = U_{Pr} - \bigcup_{i=0}^{|UI|} UI_i = U_{Pr} \cap \left(\bigcup_{i=0}^{|UI|} UI_i\right)^C = U_{Pr} \cap \bigcap_{i=0}^{|UI|} UI_i^C.$$

It has recently been demonstrated that attitude change during the decision process begins before a choice is made, not after as traditionally believed, indicating that dissonance reduction and spreading of alternatives are already occurring during the decision process. Furthermore, it appears that the person making the choice seeks

positive features that are distinctive to one of the choice alternatives (Kitayama et al., 2013; Jarcho et al., 2011). Festinger also implies that dissonance reduction begins to occur while making the choice (Festinger, 1962, pp.93 & 95; Axiom 1). These studies lead to the argument that uncertainty-based information is required to reduce dissonance caused by an ambiguous socially deviant event. Furthermore, for the queue situation, the most appealing choice appears to be non-reaction to an intrusion since it offers minimal risk of negative consequences. If that is the case, then the subject of interest should be focusing his or her uncertainty-based information search on non-reaction from the intermediate queue members, indicating legitimacy and a reduced need for confrontation with the intruder by the subject of interest. We will shortly observe that both arguments are supported by experimental data.

When the intrusion occurs, there is the initially induced dissonance and resulting need to reduce it. To reduce dissonance in ambiguous situations, and not increase it due to an error (e.g., embarrassment or offense), reliable dissonance-reducing information is needed on which to base a choice regarding whether the dissonance-causing event is legitimate or illegitimate. Before making the choice as to whether to confront the intruder, uncertainty-based information is desired to reduce the situational uncertainty. If the choice ultimately made is in error, then embarrassment (Miller, 2001) and/or justifiable confrontation may ensue. Regarding the (+1) position of interest, uncertainty-based information is limited to any immediately preceding social interaction and/or questioning the intruder as to why he or she entered the queue at that position. The initial uncertainty U_{Pr} prompts the need for information on which to base a choice of reaction or no reaction and thus reduce dissonance. It is therefore argued that U_{Pr} is independent of the uncertainty-based information sought to reduce uncertainty by the subject of interest. It may then be stated for position $(+1)$ that, with no intermediate queue members providing information,

$$P\left(R_{+1}\right) \equiv P\left(U_{Po}\right) = P\left(U_{Pr} \cap \bigcap_{i=0}^{|UI|=0} UI_i^C\right) = P\left(U_{Pr}\right)$$
$$\equiv G_{ES}\left(\overline{es}_{p,m}\right).$$

If infinite (i.e., exhaustive) uncertainty-based information is obtained before making the choice of reaction or no reaction in response to the socially deviant event, then uncertainty has been eliminated so that $U_{Po} \to 0$ as $|UI| \to \infty$. This unlikely real-world situation leads to

$$P\left(U_{Po}\right) = P\left(U_{Pr} \cap \bigcap_{i=0}^{|UI|=\infty} UI_i^C\right) = 0 \; implying \; U_{Pr} = \bigcup_{i=0}^{|UI|=\infty} UI_i.$$

Taking this approach indicates that uncertainty-based information pertains only to the uncertainty created by the social norm deviation. In other words

$$\bigcup_{i=0}^{|UI|} UI_i \subseteq U_{Pr} \; for \; |UI| \geq 0.$$

For the case where $0 < |UI| < \infty$, independent of the member of interest, uncertainty-based information is obtained from intermediate queue members immediately after the intrusion, and in the absence of attitude inoculation,

$$P(U_{Po}) = P\left(U_{Pr} \cap \bigcap_{i=0}^{|UI|} UI_i^{\,c}\right) = P(U_{Pr}) \cdot P\left(\bigcap_{i=0}^{|UI|} UI_i^{\,c}\right)$$

with the implication that $\lim_{|UI| \to \infty} P\left(\bigcap_{i=0}^{|UI|} UI_i^{C}\right) = 0.$

We already know that if $|UI| = 0 \Rightarrow UI = \{\varnothing_0\}$, where no non-reacting intermediate queue members exist between the intruder(s) and subject of interest, then in the absence of any group uncertainty-based information

$$P\left(\bigcap_{i=0}^{|UI|=0} UI_i^{C}\right) = 1.$$

To put this all together, we start with the Chain Rule,

$$P\left(\bigcap_{i=0}^{|UI|} UI_i^{C}\right) = \prod_{i=0}^{|UI|-1} P\left(UI_{|UI|-i}^{C} \mid \bigcap_{j=0}^{|UI|-(i+1)} UI_j^{C}\right).$$

Viewing each of the $|UI|$ non-reacting intermediate queue members as intensity increments of uncertainty-based information which the subject of interest uses in determining whether to react or not, and applying Equation 1.5, we have for $|UI| \geq 1$,

$$\prod_{i=0}^{|UI|-1} P\left(UI_{|UI|-i}^{C} \mid \bigcap_{j=0}^{|UI|-(i+1)} UI_j^{C}\right)$$
$$= \prod_{i=0}^{|UI|-1} P\left[k \cdot ln\left(\frac{|UI| - i + noise}{|UI| - (i+1) + noise}\right)\right].$$

In the context of this work, Melamed et al. (2019, p. 2) would imply that when an intruder becomes part of the queue group, this creates a disagreement between the initial queue members and the intruder, generating uncertainty as a result. By conforming to queue group member opinion, a subject of interest within the original queue group reduces his or her uncertainty. Referring to Festinger (1962, p. 95), the terms uncertainty and cognitive dissonance are interchangeable for the given condition. To further make this argument, FeldhamHall and Shenhav (2019, pp. 3–4) relate uncertainty with negative affective reactions. Harmon-Jones et al. (2015, p. 185) using their action-based model suggest that an individual's negative affective state of dissonance is aroused when cognitions with action implications are aroused. Maikovich (2005) uses the radicalization process in terrorist organizations to create a link between cognitive dissonance, low tolerance for uncertainty, and those who resort to

violent terrorism. Beasley (2016, pp. 7, 10–11) notes that conditions which involve uncertainty and violation of expectations (which would include social norms) are particularly prone to produce dissonance in an individual. Finally, Randles et al. (2015, p. 706) provide an interpretation of their findings indicating that dissonance may lead to uncertainty. It seems the academic community has been consistently indicating some link between cognitive dissonance and uncertainty since 1962. Additionally, uncertainly reduction begins before the choice is made, or in this case, the choice of reacting or not reacting.

What has been shown so far is that

$$P\left(R_{+|UI|+1}\right) = G_{ES}\left(\overline{es}_{p,m}\right)\cdot P\left(\bigcap_{i=0}^{|UI|} UI_i^C\right)$$
$$= G_{ES}\left(\overline{es}_{p,m}\right)\cdot \prod_{i=0}^{|UI|-1} P\left[k\cdot ln\left(\frac{|UI|-i+noise}{|UI|-(i+1)+noise}\right)\right].$$

For $UI = \left\{\varnothing_0, NR_1, NR_2, \ldots, NR_{|N|}\right\}$, $P\left[Z > z\left(|UI|\right)\right]$ for the random variable Z is defined as

$$P\left[Z > z\left(|UI|\right)\right] = \prod_{i=0}^{|UI|-1} P\left[k\cdot ln\left(\frac{|UI|-i+noise}{|UI|-(i+1)+noise}\right)\right]$$

Where the probability function $P\left[Z > z\left(|UI|\right)\right]$ is just the survival function indicating the probability a subject of interest has not yet reacted based on the amount of uncertainty-based information available at the time. For the constraints provided, and the derivation of $P\left[Z > z\left(|UI|\right)\right]$, application of the negative exponential (Efthimiou, 2010, p. 83) for $P\left[Z > z\left(|UI|\right)\right]$ is appropriate. As an example, using $|UI| = 2$ and the function $z\left(|UI|\right)$ when applied to the standard negative exponential,

$$G_{ES}\left(\overline{es}_{p,m}\right)\cdot e^{-z(2)} = G_{ES}\left(\overline{es}_{p,m}\right)\cdot e^{-k\cdot ln\left(\frac{2+noise}{1+noise}\right)}\cdot e^{-k\cdot ln\left(\frac{1+noise}{noise}\right)}$$
$$= G_{ES}\left(\overline{es}_{p,m}\right)\cdot e^{-k\cdot\left[ln\left(\frac{2+noise}{1+noise}\right)+ln\left(\frac{1+noise}{noise}\right)\right]}$$
$$= G_{ES}\left(\overline{es}_{p,m}\right)\cdot e^{-k\cdot\left[ln\left(\frac{2+noise}{noise}\right)\right]}.$$

Constraints may be shown to have been met using the example $G_{ES}\left(\overline{es}_{p,m}\right)\cdot e^{-z(2)}$ and a proof via induction, assuming $0 < noise < \infty$, so that for both $k > 0$ *and* $0 < noise < \infty$,

$$\lim_{|UI|\to\infty} G_{ES}\left(\overline{es}_{p,m}\right)\cdot e^{-k\cdot\left[ln\left(\frac{|UI|+noise}{noise}\right)\right]}$$

$$= 0 \le G_{ES}\left(\overline{es}_{p,m}\right)\cdot e^{-k\cdot\left[ln\left(\frac{|UI|+noise}{noise}\right)\right]} \le G_{ES}\left(\overline{es}_{p,m}\right).$$

Accounting for the independence of certain random variables, this allows derivation of probability of reaction for some specified uncertainty-based information set $UI = \left\{\varnothing_0, NR_1, NR_2, \ldots, NR_{|N|}\right\}, |UI| \ge 0$, and given values $\overline{es}$, p, *and* m, such that

$$P\left[ES \le \overline{es}_{p,m}, Z > z\left(|UI|\right)\right] = G_{ES}\left(\overline{es}_{p,m}\right)\cdot e^{-z\left(|UI|\right)} = \left(1 - e^{-\overline{es}_{p,m}}\right)\cdot e^{-z\left(|UI|\right)} \qquad \textit{Equation } 3.2$$

With

$$z\left(|UI|\right) = \sum_{i=0}^{|UI|-1} k\cdot ln\left(\frac{|UI|-i+noise}{|UI|-(i+1)+noise}\right) \textit{ for } |UI| \ge 1,$$

$$\textit{else } z\left(|UI|\right) = 0.$$

The random variables *ES and Z* are independent if the number of members behind the position of interest and do not depend on $|UI|$. With that, we now have the tools to derive the variables *k* and *noise* for the function $z\left(|UI|\right)$.

Using data from Table 3.6, we may now evaluate Condition 2 using Equation 3.2. Using $\overline{es}_{L,1} = 0.114$ and $\overline{es}_{H,1} = 0.456$, as derived from Schmitt et al. (1992) and

TABLE 3.6 Condition 2 Empirical Data

Q(P|N|M) = Q(P intruders | N queue members between intruder and position of interest | M queue members behind queue position of interest) = Position Probability of Occurrence				
Position (-1)	Position (-2)	Position (-3)	Position (-4)	Front of Queue
3/24	**0/22**	**0/18**	**0/3**	# React /# Opportunities
1 Intruder (Condition 2)	**Notes**: 1) Appendix A has k=2 and l_{100}=62.7: theoretical M = 1.4 ≈ documented M_{mean} = 1.2 in the Q(1|1|*) configuration : No extrapolation for "***M***" necessary. 2) The original 1978 summary data and two of the graduate student reports indicate 0/8 at (+4) versus 0/9 per Milgram et al (1986). Milgram et al (1986) takes precedence.			
Position (+1)	Position (+2)	Position (+3)	Position (+4)	Position (+5)
0/24	**4/24**	**0/15**	**0/9**	**0/5**
Confederate (Graduate Student) Non-reactive Buffer	***Q*(1|1|3) = 5/24**	***Q*(1|2|2) = 5/15**	***Q*(1|3|1) = 5/9**	***Q*(1|4|0) = 5/5**
	***Q*(1|1|2) = 4/24**	***Q*(1|2|1) = 4/15**	***Q*(1|3|0) = 4/9**	Milgram et al (1986)
	***Q*(1|1|1) = 6/24**	***Q*(1|2|0) = 6/15**	supplemented by unpublished data obtained from	
	***Q*(1|1|0) = 9/24**	The Stanley Milgram Papers (MS 1406), Series No. IV, Box No. 100, Folder No. 131		
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Condition 1: (+1) of Milgram et al. (1986), we begin by deriving $G_{ES}\left(\overline{es}_{I,1,m}\right)\cdot e^{-z(1)}$ for Condition 2 position (+2). The derivation process is the same as for Conditions 1 and 4 at position (+1) using the Law of Total Probability while incorporating Equation 3.2 with $UI=\{NR_{+1}\}\Rightarrow|UI|=1\ and\ P(UI=\{NR_{+1}\})=1$. It is worth repeating due to lack of experimental raw data, homogeneity with regard to m of high (H) and low (L) unit stimulus sensation magnitude queues must be assumed.

$$P\left[ES\le\overline{es}_{I,1,m},\overline{es}_{H},Q(1|1|0)\right]\cdot P\left[Z>z(|UI|)\right]$$
$$=\left(1-e^{-.456}\right)\cdot\left(\frac{19\,\overline{es}_{H}\ Trials}{24\ Trials}\right)\cdot\left(\frac{9}{24}\right)\cdot e^{-z(1)}$$
$$=0.109\cdot e^{-z(1)}$$

$$P\left[ES\le\overline{es}_{I,1,m},\overline{es}_{H},Q(1|1|1)\right]\cdot P\left[Z>z(|UI|)\right]$$
$$=\left(1-e^{-0.456\cdot ln\left(e\cdot\frac{1+1.147}{1.147}\right)}\right)\cdot\left(\frac{19}{24}\right)\cdot\left(\frac{6}{24}\right)\cdot e^{-z(1)}$$
$$=0.104\cdot e^{-z(1)}$$

$$P\left[ES\le\overline{es}_{I,1,m},\overline{es}_{H},Q(1|1|2)\right]\cdot P\left[Z>z(|UI|)\right]$$
$$=\left(1-e^{-0.456\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)}\right)\cdot\left(\frac{19}{24}\right)\cdot\left(\frac{4}{24}\right)\cdot e^{-z(1)}$$
$$=0.079\cdot e^{-z(1)}$$

$$P\left[ES\le\overline{es}_{I,1,m},\overline{es}_{H},Q(1|1|3)\right]\cdot P\left[Z>z(|UI|)\right]$$
$$=\left(1-e^{-0.456\cdot ln\left(e\cdot\frac{3+1.147}{1.147}\right)}\right)\cdot\left(\frac{19}{24}\right)\cdot\left(\frac{5}{24}\right)\cdot e^{-z(1)}$$
$$=0.107\cdot e^{-z(1)}$$

$$P\left[ES\le\overline{es}_{I,1,m},\overline{es}_{L},Q(1|1|0)\right]\cdot P\left[Z>z(|UI|)\right]$$
$$=\left(1-e^{-0.114}\right)\cdot\left(\frac{5\,\overline{es}_{L}\ Trials}{24\ Trials}\right)\cdot\left(\frac{9}{24}\right)\cdot e^{-z(1)}$$
$$=0.008\cdot e^{-z(1)}$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_L, Q\left(1|1|1\right)\right] \cdot P\left[Z > z\left(|UI|\right)\right]$$
$$= \left(1 - e^{-0.114 \cdot ln\left(e \cdot \frac{1+1.147}{1.147} \right)} \right) \cdot \left(\frac{5}{24} \right) \cdot \left(\frac{6}{24} \right) \cdot e^{-z(1)}$$
$$= 0.009 \cdot e^{-z(1)}$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_L, Q\left(1|1|2\right)\right] \cdot P\left[Z > z\left(|UI|\right)\right]$$
$$= \left(1 - e^{-0.114 \cdot ln\left(e \cdot \frac{2+1.147}{1.147} \right)} \right) \cdot \left(\frac{5}{24} \right) \cdot \left(\frac{4}{24} \right) \cdot e^{-z(1)}$$
$$= 0.007 \cdot e^{-z(1)}$$

$$P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_L, Q\left(1|1|3\right)\right] \cdot P\left[Z > z\left(|UI|\right)\right]$$
$$= \left(1 - e^{-0.114 \cdot ln\left(e \cdot \frac{3+1.147}{1.147} \right)} \right) \cdot \left(\frac{5}{24} \right) \cdot \left(\frac{5}{24} \right) \cdot e^{-z(1)}$$
$$= 0.010 \cdot e^{-z(1)}$$

Accounting for round-off error in the aforementioned and as before assuming $P\left[\overline{es}_I \mid Q\left(1|0|m\right)\right] = P\left[\overline{es}_I\right]$,

$$P\left(R_{+|UI|+1}\right) = P\left(R_{+2}\right)$$
$$= \sum_{I \in \{H,L\}} \sum_{m \in M} P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_I, Q\left(1|1|m\right)\right] \cdot P\left[Z > z\left(|UI|\right)\right]$$
$$= \sum_{I \in \{H,L\}} \sum_{m \in M} G_{ES}\left(\overline{es}_{I,1,\{NR_1\},m}\right) \cdot P\left(\overline{es}_I\right) \cdot P\left[Q\left(1|1|m\right)\right] \cdot e^{-z(1)}$$
$$= 0.433 \cdot e^{-z(1)}$$
$$= \frac{4}{24} = 0.167 \left(\textit{empirical result}\right).$$

$$e^{-z(1)} = \frac{0.167}{0.433} = 0.386, or\ z\left(1\right) = -ln\left(0.386\right) = 0.952.$$

TABLE 3.7 Condition 5 Empirical Data

$Q(P\|N\|M) = Q(P$ *intruders* $\mid N$ *queue members between intruder and position of interest* $\mid$ M *queue members behind queue position of interest*$) =$ *Position Probability of Occurrence*				
Position (-1)	Position (-2)	Position (-3)	Position (-4)	Front of Queue
2/20	**0/18**	**0/13**	**0/7**	# React /# Opportunities
2 Intruders (Condition 5)	**Notes**: 1) Appendix A has k=10 and l_{100}=61.1: theoretical M = 1.1 ≈ documented M_{mean} = 1.0 in the Q(2|1|*) configuration: No extrapolation for "M" necessary. 2) Original 1978 summary data and two graduate reports indicate 5/20 for position (+2) versus 4/20 as reported in Milgram et al (1986). Milgram et al (1986) takes precedence.			
Position (+1)	Position (+2)	Position (+3)	Position (+4)	Position (+5)
0/20	**4/20**	**0/15**	**0/4**	**0/0**
Confederate (Graduate Student) Non-reactive Buffer	$Q(2\|1\|3) = 0/20$	$Q(2\|2\|2) = 0/15$	$Q(2\|3\|1) = 0/4$	No Opportunties
	$Q(2\|1\|2) = 4/20$	$Q(2\|2\|1) = 4/15$	$Q(2\|3\|0) = 4/4$	Milgram et al (1986)
	$Q(2\|1\|1) = 11/20$	$Q(2\|2\|0) = 11/15$	supplemented by unpublished data obtained from	
	$Q(2\|1\|0) = 5/20$	The Stanley Milgram Papers (MS 1406), Series No. IV, Box No. 100, Folder No. 131		
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With $z(1)$ derived, we may now evaluate Condition 5, which also has a single buffer as Condition 2, and check for repeatability using a queue experiencing two simultaneous intruders.

Applying $z(1) = 0.952$ as derived, in the same manner as Condition 2, it is found by spreadsheet calculations that $\overline{es}_{H,2} = 0.780$ and $\overline{es}_{L,2} = 0.195$ when the total probability of reaction equals $\frac{4}{20}$ (*i.e.,* 0.200) for Condition 5: (+2), see Table 3.7. The sample mean of $\overline{es}_{H,2} = 0.780$ is a feasible solution with Conditions 1 and 2 as per Corollary 2.3. We can now move to Condition 6 and get an approximate value for z(2). If $z(1)$ and $z(2)$ meet Corollary 2.3 requirements, then Theorem 2.5 may be applied to calculate a noise value related to member interpretation of intermediate member reactions. Table 3.8 summarizes the necessary Condition 6 reaction data as a function of position.

The same approach as Condition 2 and 5 is used for Condition 6: (+3) but using $\overline{es}_{H,2} = 0.780$ (note: there are no low sensation magnitude queues in this condition). Given $UI = \{NR_{+1}, NR_{+2}\}$ and that $P(UI) = 1$, the result is,

$$P(R_{+3}) = \sum_{I \in \{H\}} \sum_{m \in M} P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_I, Q(2|2|m)\right]$$

$$\cdot P\left[Z > z(|UI|)\right] = \sum_{I \in \{H\}} \sum_{m \in M} G_{ES}\left(\overline{es}_{I,2,m}\right)$$

$$\cdot P\left(\overline{es}_I\right) \cdot P\left[Q(2|2|m)\right] \cdot e^{-z(2)} = 0.731 \cdot e^{-z(2)}$$

$$= \frac{3}{20} = 0.150 \left(\textit{empirical result}\right).$$

$$\textit{As before, } e^{-z(2)} = \frac{0.150}{0.731} = 0.205, \textit{or } z(2)$$

$$= -ln(0.205) = 1.585.$$

TABLE 3.8 Condition 6 Empirical Data

$Q(P\|N\|M) = Q(P$ *intruders* $\mid N$ *queue members between intruder and position of interest* $\mid M$ *queue members behind queue position of interest*$) =$ *Position Probability of Occurrence*				
Position (-1)	Position (-2)	Position (-3)	Position (-4)	Front of Queue
2/20	**0/18**	**0/18**	**0/13**	# React /# Opportunities
2 Intruders (Condition 6)	**Notes**: 1) Appendix A has k=10 and l_{100}=76.3: theoretical M = 2.1 slightly > documented M_{mean} = 1.3 in the Q(2|2|*) configuration: No extrapolation for "M" necessary. 2) Original 1978 summary data and two graduate reports indicate 2/16 for position (+4) versus 2/17 as reported in Milgram et al (1986). Milgram et al (1986) takes precedence.			
Position (+1)	Position (+2)	Position (+3)	Position (+4)	Position (+5)
0/20	**0/20**	**3/20**	**2/17**	**0/10**
Confederate (Graduate Student) Non-reactive Buffer	**Confederate (Graduate Student) Non-reactive Buffer**	$Q(2\|2\|2) = 10/20$	$Q(2\|3\|1) = 10/17$	$Q(2\|4\|0) = 10/20$
		$Q(2\|2\|1) = 7/20$	$Q(2\|3\|0) = 7/17$	Milgram et al (1986)
		$Q(2\|2\|0) = 3/20$	supplemented by unpublished data obtained from	
		The Stanley Milgram Papers (MS 1406), Series No. IV, Box No. 100, Folder No. 131		
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It may also be observed that Corollary 2.3 applies since $1 < \frac{z(2)}{z(1)} = \frac{1.585}{0.952} < 2$.

As expected, a logarithmic approach to modeling presents itself as predicted from the derivation of $z(|N|)$ by noting that for two confederate nonreactive buffers,

$$z(2) = 1.585 \cong z(1) + z(1) \cdot ln(2) = 0.952 \cdot ln(e \cdot 2) = 1.612.$$

Hence, a feasible value exists for a noise term from the data related to a member's ability to observe and interpret the non-reactions of other group members closer to the dissonant-causing event. In this case, non-reaction by a queue member immediately behind the intrusion point provides information to other queue members. Non-reaction implies to others that the intrusion did not appear to be a significant deviation of group norms and therefore is likely to be of low-sensation magnitude. The member immediately following may use this information to reduce his or her uncertainty as to what occurred and thereby determine whether to react or not react.

In the example of two nonreactive queue members immediately behind the intrusion point, the amount of uncertainty as measured in probability space by queue members in subsequent positions is about one-fifth (i.e., $e^{-z(2)} = 0.205$) of the initial amount. The uncertainty-based information noise variable N_{UI} may be noise interfering with obtaining uncertainty-based information, or it may be a constant. If noise increases, the ability of the member of interest to interpret and make use of information through observation of pertinent members decreases. Described in this way, it would seem N_{UI} and N_{UA} should be positively correlated, given members in attractive (i.e., cohesive) groups would be better able to interpret one another's reactions toward any deviation of group social norms.

Given the values for $z(1)$ and $z(2)$ meet Corollary 2.3 requirements, Theorem 2.4 and Theorem 2.5 may be applied to find $\widehat{N}_{UI}$ and $\overline{es}(\widehat{N}_{UI})$, respectively. The value of $\widehat{N}_{UI}$ supporting,

$$\left[0.952-\frac{1.585\cdot ln\left(\frac{1+\widehat{N}_{UI}}{\widehat{N}_{UI}}\right)}{ln\left(\frac{2+\widehat{N}_{UI}}{\widehat{N}_{UI}}\right)}\right]+\left[1.585-\frac{0.952\cdot ln\left(\frac{2+\widehat{N}_{UI}}{\widehat{N}_{UI}}\right)}{ln\left(\frac{1+\widehat{N}_{UI}}{\widehat{N}_{UI}}\right)}\right]$$
$$=0$$

is $\widehat{N}_{UI}=1.553$. Using $\widehat{N}_{UI}=1.553$ with Theorem 2.4 results in

$$\overline{es}\left(\widehat{N}_{UI}\right)=\frac{1}{2}\cdot\left[\frac{0.952}{ln\left(\frac{1+\widehat{N}_{UI}}{\widehat{N}_{UI}}\right)}+\frac{1.585}{ln\left(\frac{2+\widehat{N}_{UI}}{\widehat{N}_{UI}}\right)}\right]=1.195.$$

From this, we have

$$\overline{z}\left(1\right)=1.915\cdot ln\left(\frac{1+1.553}{1.553}\right)=0.952;$$

$$\overline{z}\left(2\right)=1.915\cdot ln\left(\frac{2+1.553}{1.553}\right)=1.585.$$

Stated in probability space, $e^{-\overline{z}(1)}=e^{-0.952}=0.386$, *or* $1-e^{-\overline{z}(1)}=0.614$.

The probability 0.614 is extremely close to a very familiar value. The value 0.618 is the inverse of the Golden Ratio. In operations research, the value 0.618 is used in the Golden Section search to optimize speed of finding the maximum value of a unimodal function (Winston, 1987, pp. 545–549) with minimal computational effort. Referring to Tanner et al. (1967, p. 352), based on a least squares fit of the data, probability of the observers identifying the sound as the target sound approaches the inverse Golden Ratio as the probability of the target sound being introduced approaches 1. Benjafield and Adams-Weber (1976) provide a hypothesis leading to even more interesting and potentially pertinent results. Let positive events be associated with a non-reaction indicating acceptable behavior by group members, and let negative events be associated with a reaction to indicate unacceptable behavior under those same norms. Benjafield and Adams-Weber (1976, pp. 13–14) hypothesize that socially deviant behaviors become most noticeable when compared to the acceptable background of non-deviant behaviors when the proportion of negative events (social deviation) divided by the proportion of positive events (group-accepted behavior) tends to the Golden Section value of 0.618.

In summary, for $|UI|$ nonreactive buffers between the intruder(s) and the $(+|UI|+1)$ position of interest, $e^{-z\left(|UI|\right)}$ may be expressed for the New York queue as

TABLE 3.9 Condition 3 Empirical Data

$Q(P\|N\|M) = Q(P$ *intruders* $\mid N$ *queue members between intruder and position of interest* $\mid$ M *queue members behind queue position of interest*$) =$ *Position Probability of Occurrence*				
Position (-1)	Position (-2)	Position (-3)	Position (-4)	Front of Queue
1/20	**0/18**	**0/17**	**0/9**	# React /# Opportunities
1 Intruder (Condition 3)	**Notes**: 1) Appendix A has k=10 and l_{100}=74.7: theoretical M = 2 > documented M_{mean} = 1.4 in the Q(1|2|*) configuration: No difference though since current Q(1|2|2) = theoretical Q(1|2|2): No extrapolation for "***M***" necessary.			
Position (+1)	Position (+2)	Position (+3)	Position (+4)	Position (+5)
0/20	**0/20**	**0/20**	**0/18**	**0/9**
Confederate (Graduate Student) Non-reactive Buffer	**Confederate (Graduate Student) Non-reactive Buffer**	$Q(1\|2\|2) = 9/20$	$Q(1\|3\|1) = 9/18$	$Q(1\|4\|0) = 9/9$
		$Q(1\|2\|1) = 9/20$	$Q(1\|3\|0) = 9/18$	Milgram et al (1986)
		$Q(1\|2\|0) = 2/20$	supplemented by unpublished data obtained from	
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$$e^{-\bar{z}(|UI|)} = e^{-1.915 \cdot ln\left(\frac{|UI|+1.553}{1.553}\right)} \qquad \textit{Equation } 3.3$$

where all elements in UI are nonreacting.

With Equation 3.3 available, Condition 3 data shown in Table 3.9 may now be analyzed.

Using $\overline{es}_{H,1} = 0.456$ and $\overline{es}_{L,1} = 0.114$ from Condition 1, and applying the same process as demonstrated for Conditions 1 and 6 while using values from Table 3.9, we obtain,

$$\begin{aligned} P(R_{+3}) &= \sum_{I=\{H,L\}} \sum_{m \in M} P\left[ES \le \overline{es}_{I,1,m}, \overline{es}_I, Q(1|2|m)\right] \cdot P\left[Z > z(|UI|)\right] \\ &= \sum_{I=\{H,L\}} \sum_{m \in M} G_{ES}\left(\overline{es}_{I,1,m}\right) \cdot P\left[\overline{es}_I\right] \cdot P\left[Q(1|2|m)\right] \cdot e^{-\bar{z}(2)} \\ &= 0.063. \end{aligned}$$

The result of 0.063 for Condition 3: (+3) is well within the 95 percent credibility interval of (0.0, 0.133). We now have all the mathematical tools, when combined with Equation 3.3, to analyze empirical data for Conditions 1 and 4 at the (+1) positions and for Conditions 2, 3, 5, and 6 at the positions just after their respective nonreactive buffers. To solve for the remaining Conditions and positions requires addressing those intermediate members who react between the intruder(s) and position of interest.

3.3.4 Impact of Reacting Members: Deriving y(\|UI\|) Using Condition 4: (+2)

Experiments and data collection are difficult, and the intrusion experiment by Stanley Milgram and his graduate students in 1978 was even more so. Though the experiment and the data are nothing less than outstanding, there are always lessons learned that

should be documented for others to build on. It is also important to understand the data. To this end, in his 1978 student report, graduate student David Niemeroff wrote the following (Stanley Milgram Papers (MS1406), with copyright permission from the Estate of Alexandra Milgram):

> *We can observe with 2 intruders and no buffers nearly all the lines objected. Interestingly, in this condition the only two lines that did not object were the two information booth lines at Grand Central station*.

This clearly refers to Condition 4 since there are two intruders and no buffers. David Niemeroff further notes that these two lines were the fastest and also the most amorphous, alternating their topology every minute or so between that of a straight line or a cluster as is sometimes seen at a bus stop (Stanley Milgram Papers, MS 1406, Box 101, Folder 136, pp. 8–9). This indicates the ambiguity of queue position status. It also indicates potential confusion of roles and responsibilities for maintaining group boundaries separating those in queue from those not in queue. The importance is that being amorphous leads to position uncertainty, social uncertainty (embarrassment if wrong to react), and greater uncertainty of attraction. It is suggested here that the information booth was not representative when viewed within the context of all the other queue locations detailed in Table 3.3.

In 1978, when preparing for the experiment, graduate student Joyce Wackenhut defined the queue in her planning notes as a clearly defined social structure. In his translation of Parsons (1951), Mann (1969, p. 349) provides the properties of a social system of which he states the queue is an embryonic representative. He states that three properties of any social system are: (1) two or more members occupying differentiated statuses or positions and performing differentiated roles; (2) some organized pattern governing the relationships of the members, describing their rights and obligations with respect to one another; and (3) some set of common norms and values, together with various types of shared cultural objects and symbols. As a lesson learned, queues like the information booth queue as described by David Niemeroff are not well-differentiated or defined social systems and should be avoided for queue analysis.

Condition 4: (+1) has the most reactions (20 out of 23) of any (+1) position for the six conditions. This makes it an excellent candidate to derive $y(1)$ using Condition 4: (+2) results by replacing $z(1)$ for $UI = \{NR_{+1}\}$ with $y(1)$ for $UI = \{R_{+1}\}$ in Equation 3.2. Given David Niemeroff's report, it is apparent there are 21 observations of interest here, and of those 21, 20 had reactions at the (+1) position. Of those 21 observations, there were 10 reactions at the (+2) position. With that simple observation, we can guess up front that $\boldsymbol{e}^{-\bar{y}(1)}$ will be somewhere around 0.5. Ultimately, once $\boldsymbol{e}^{-\bar{y}(n)}$ is obtained in a mathematically traceable manner, all remaining Conditions at positions beyond (+1) may be theoretically calculated using Equation 3.2 where the set UI now may contain any ranked combination of reacting and non-reacting members.

Using the Law of Total Probability with Equation 3.2, while now accounting for the possibility of a reactive or nonreactive intermediate queue member, and assuming as before, based on lack of available information that $P\left[\overline{es}_I \mid Q\left(2|1|m\right)\right] = P\left(\overline{es}_I\right)$, results in,

$$P\left(R_{+2}\right)=\sum_{r\in\left\{R_{+1},NR_{+1}\right\}}\sum_{I\in\left\{H,L\right\}}\sum_{m\in M}G_{ES}\left(\overline{es}_{I,2,m}\right)\cdot P\left(\overline{es}_{I}\right)\cdot P\left[Q\left(2\middle|1\middle|m\right)\right]\cdot H_{+2}\left(r\right)\cdot P\left(r\right),$$

$$\textit{defining } H_{+2}\left(r\right)\cdot P\left(r\right)=e^{-\bar{y}(1)}\cdot G\left(\overline{es}_{I,2,m+1}\right)$$

$$\textit{if } r=R_{+1} \textit{ else } H_{+2}\left(r\right)\cdot P\left(r\right)$$

$$=e^{-\bar{z}(1)}\left[1-G\left(\overline{es}_{I,2,m+1}\right)\right],\ \textit{if } r=\mathrm{NR}_{+1}.$$

Knowing $\overline{es}_{H,2}=0.780$, but for this situation using $\overline{es}_{H,2}=1.005$ as derived for Condition 4: (+1) to solve for $e^{-y(1)}$ under the same conditions, we may now solve for $e^{-y(1)}$ in the following manner using Condition 4: (+2):

$$\sum_{r\in\left\{R_{+1},NR_{+1}\right\}}P\left[ES\le\overline{es}_{I,2,m},\overline{es}_{H},Q\left(2\middle|1\middle|0\right)\right]\cdot H_{+2}\left(r\right)\cdot P(r)$$

$$=\left(1-e^{-1.005}\right)\cdot\left(\frac{19\ \overline{es}_{H}\ Trials}{23\ Trials}\right)\cdot\left(\frac{1}{23}\right)$$

$$\cdot\left[e^{-\bar{z}(1)}\cdot\left[1-G\left(\overline{es}_{H,2,m}\right)\right]+e^{-\bar{y}(1)}\cdot G\left(\overline{es}_{H,2,m}\right)\right]$$

$$=0.0228\cdot\left[0.386\cdot e^{-1.005\cdot ln\left(e\cdot\frac{1+1.147}{1.147}\right)}+e^{-y(1)}\cdot\left(1-e^{-1.005\cdot ln\left(e\cdot\frac{1+1.147}{1.147}\right)}\right)\right]=0.0017+0.0183\cdot e^{-y(1)}$$

$$\sum_{r\in\left\{R_{+1},NR_{+1}\right\}}P\left[ES\le\overline{es}_{I,2,m},\overline{es}_{H},Q\left(2\middle|1\middle|1\right)\right]\cdot H_{+2}\left(r\right)\cdot P(r)$$

$$=\left(1-e^{-1.005\cdot ln\left(e\cdot\frac{1+1.147}{1.147}\right)}\right)\cdot\left(\frac{19}{23}\right)\cdot\left(\frac{2}{23}\right)$$

$$\cdot\left[e^{-\bar{z}(1)}\cdot\left[1-G\left(\overline{es}_{H,2,m}\right)\right]+e^{-y(1)}\cdot G\left(\overline{es}_{H,2,m}\right)\right]$$

$$=0.0578\cdot\left[0.386\cdot e^{-1.005\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)}+e^{-y(1)}\cdot\left(1-e^{-1.005\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)}\right)\right]$$

$$=0.003+0.050\cdot e^{-y(1)}$$

$$\sum_{r\in\{R_{+1},NR_{+1}\}} P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_H, Q\left(2|1|2\right)\right]\cdot H_{+2}\left(r\right)\cdot P(r)$$

$$= \left(1-e^{-1.005\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)}\right)\cdot\left(\frac{19}{23}\right)\cdot\left(\frac{3}{23}\right)$$

$$\cdot\left[e^{-\overline{z}(1)}\cdot\left[1-G\left(\overline{es}_{H,2,m}\right)\right]+e^{-\overline{y}(1)}\cdot G\left(\overline{es}_{H,2,m}\right)\right]$$

$$= 0.0934\cdot\left[0.386\cdot e^{-1.005\cdot ln\left(e\cdot\frac{3+1.147}{1.147}\right)}+e^{-y(1)}\cdot\left(1-e^{-1.005\cdot ln\left(e\cdot\frac{3+1.147}{1.147}\right)}\right)\right]$$

$$= 0.0036+0.0840\cdot e^{-y(1)}$$

$$\sum_{r\in\{R_{+1},NR_{+1}\}} P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_H, Q\left(2|1|10.3\right)\right]\cdot H_{+2}\left(r\right)\cdot P(r)$$

$$= \left(1-e^{-1.005\cdot ln\left(e\cdot\frac{10.3+1.147}{1.147}\right)}\right)\cdot\left(\frac{19}{23}\right)\cdot\left(\frac{17}{23}\right)$$

$$\cdot\left[e^{-\overline{z}(1)}\cdot\left[1-G\left(\overline{es}_{H,2,m}\right)\right]+e^{-\overline{y}(1)}\cdot G\left(\overline{es}_{H,2,m}\right)\right]$$

$$= 0.588\cdot\left[0.386\cdot e^{-1.005\cdot ln\left(e\cdot\frac{11.3+1.147}{1.147}\right)}+e^{-y(1)}\cdot\left(1-e^{-1.005\cdot ln\left(e\cdot\frac{11.3+1.147}{1.147}\right)}\right)\right]$$

$$= 0.0076+0.5688\cdot e^{-y(1)}$$

$$\sum_{UI\in\{R_{+1},NR_{+1}\}} P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_L, Q\left(2|1|0\right)\right]\cdot H_{+2}\left(r\right)\cdot P(r)$$

$$= \left(1-e^{-0.251}\right)\cdot\left(\frac{4\,\overline{es}_L\ Trials}{23 Trials}\right)\cdot\left(\frac{1}{23}\right)\cdot$$

$$\cdot\left[e^{-\overline{z}(1)}\cdot\left[1-G\left(\overline{es}_{H,2,m}\right)\right]+e^{-\overline{y}(1)}\cdot G\left(\overline{es}_{H,2,m}\right)\right]$$

$$= 0.0017\cdot\left[0.386\cdot e^{-0.251\cdot ln\left(e\cdot\frac{1+1.147}{1.147}\right)}+e^{-y(1)}\cdot\left(1-e^{-0.251\cdot ln\left(e\cdot\frac{1+1.147}{1.147}\right)}\right)\right]$$

$$= 0.0004+0.0006\cdot e^{-y(1)}$$

$$\sum_{UI\in\{R_{+1},NR_{+1}\}} P\left[ES \le \overline{es}_{I,2.m}, \overline{es}_L, Q\left(2|1|1\right)\right]\cdot H_{+2}\left(r\right)\cdot P(r)$$

$$= \left(1-e^{-0.251\cdot ln\left(e\cdot\frac{1+1.147}{1.147}\right)}\right)\cdot\left(\frac{4}{23\,Trials}\right)\cdot\left(\frac{2}{23}\right)$$

$$\cdot\left[e^{-\overline{z}(1)}\cdot\left[1-G\left(\overline{es}_{H,2,m}\right)\right]+e^{-\overline{y}(1)}\cdot G\left(\overline{es}_{H,2,m}\right)\right]$$

$$= 0.0051\cdot\left[0.386\cdot e^{-0.251\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)}+e^{-y(1)}\right.$$

$$\left.\cdot\left(1-e^{-0.251\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)}\right)\right]$$

$$= 0.0012+0.0020\cdot e^{-y(1)}$$

$$\sum_{UI\in\{R_{+1},NR_{+1}\}} P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_L, Q\left(2|1|2\right)\right]\cdot H_{+2}\left(r\right)\cdot P(r)$$

$$= \left(1-e^{-0.251\cdot ln\left(e\cdot\frac{2+1.147}{1.147}\right)}\right)\cdot\left(\frac{4}{23Trials}\right)\cdot\left(\frac{3}{23}\right)$$

$$\cdot\left[e^{-\overline{z}(1)}\cdot\left[1-G\left(\overline{es}_{H,2,m}\right)\right]+e^{-\overline{y}(1)}\cdot G\left(\overline{es}_{H,2,m}\right)\right]$$

$$= 0.009\cdot\left[0.386\cdot e^{-0.251\cdot ln\left(e\cdot\frac{3+1.147}{1.147}\right)}+e^{-y(1)}\right.$$

$$\left.\cdot\left(1-e^{-0.251\cdot ln\left(e\cdot\frac{3+1.147}{1.147}\right)}\right)\right]$$

$$= 0.002+0.0039\cdot e^{-y(1)}$$

$$\sum_{UI\in\{R_{+1},NR_{+1}\}} P\left[ES \le \overline{es}_{I,2,m}, \overline{es}_L, Q\left(2|1|10.3\right)\right]\cdot H_{+2}\left(r\right)\cdot P(r)$$

$$= \left(1-e^{-0.251\cdot ln\left(e\cdot\frac{10.3+1.147}{1.147}\right)}\right)\cdot\left(\frac{4}{23\,Trials}\right)\cdot\left(\frac{17}{23}\right)$$

$$\cdot\left[e^{-\overline{z}(1)}\cdot\left[1-G\left(\overline{es}_{H,2,m}\right)\right]+e^{-\overline{y}(1)}\cdot G\left(\overline{es}_{H,2,m}\right)\right]=0.0724$$

$$\cdot\left[0.386\cdot e^{-0.251\cdot ln\left(e\cdot\frac{11.3+1.147}{1.147}\right)}+e^{-y(1)}\cdot\left(1-e^{-0.251\cdot ln\left(e\cdot\frac{11.3+1.147}{1.147}\right)}\right)\right]$$

$$= 0.0120+0.0414\cdot e^{-y(1)}$$

Now add the terms just calculated and equate the sum to the empirical result of 0.435 representing the Condition 4: (+2) empirical probability of reaction to obtain,

$$0.0314+0.7693\cdot e^{-y(1)}=0.435,\ or\ e^{-y(1)}=0.5246$$

Which solves to $y(1) = 0.645$. For now, making the rather large assumption that noise, assuming that is what it is, caused by uncertainty of interpretation is the same as that for $z(1)$ given both are in the same social situation, we obtain,

$$\overline{y}\left(|UI|\right)=\frac{0.645}{ln\left(\frac{1+1.553}{1.553}\right)}\cdot ln\left(\frac{|UI|+1.553}{1.553}\right)$$

$$=1.298\cdot ln\left(\frac{|UI|+1.553}{1.553}\right),\ for\ UI$$

$$=\left\{R_n,R_{n-1},\ldots,R_1,\varnothing\right\},\ n\in\mathbb{Z}^+.$$

Equation 3.4

Like the value obtained for $e^{-\overline{z}(1)}$, the value obtained for $e^{-\overline{y}(1)}$ has potential significance as well. In the queue, the binary information obtained is either a reaction directed at the intruder or non-reaction. In information theory, the amount of uncertainty-based information obtained is maximized (highest entropy) when the probability of either binary event is 0.5 (Stone, 2015, pp. 33–38). Interpreting this, it may be that as a species, we minimize effort to confirm that everything is normal socially but maximize information intake when a confirmed social deviation event occurs. It seems that a significant amount of information theory is woven into this process, and

it is felt that those who have expertise in this area would likely find further investigation fruitful. It is also possible, given the values derived for $z(1)$ and y(1) and their potential interpretation, that $\widehat{N}_{UI}$ may be a constant, and a very interesting one if with more data it is found that

$$\frac{1+\widehat{N}_{UI}}{\widehat{N}_{UI}} \to 1.618 \text{ since } \frac{1+1.553}{1.553} = 1.644 \text{ is rather close to the Goldern Ratio already.}$$

As has been observed in the intrusion data from this chapter, some of the intermediate queue members react, while others do not. This fact requires a generalization which may be found by considering $n = |UI| \in \mathbb{Z}^0$ and the set $UI = \{\varnothing_0, UI_1, UI_2, \ldots, UI_n\}$ where each $UI_{i+1} \in \{R_{i+1}, NR_{i+1}\}$ with $i \in \mathbb{Z}^0$, $0 \le i \le |UI| - 1$ so that

$$H_{|UI|+1}(UI) = \prod_{i=0}^{|UI|-1} p\left[k(UI_{i+1}) \cdot ln\left(\frac{|UI| - i + noise}{|UI| - (i+1) + noise}\right)\right],$$

recalling $|\varnothing| = 0$.

This is equivalent to,

$$H_{|UI|+1}(UI) = e^{-\sum_{i=o}^{|UI|-1} k(UI_{i+1}) \cdot ln\left(\frac{|UI| - i + noise}{|UI| - (i+1) + noise}\right)}$$

Equation 3.5

for $|UI| \ge 1$, *and* $H_{+1}(\varnothing) = 1$, *with* $k(R_{i+1}) = 1.298$ *and* $k(NR_{i+1}) = 1.915$.

Using Equation 3.5, $UI = \{\varnothing_0, UI_1, UI_2, \ldots, UI_n\}$, $UI \in \{R_{+i}, NR_{+i}\}$, and m members behind the subject of interest having $|UI|$ intermediate queue members between him or her and the intruder(s), define,

$$P\left(R_{+|UI|+1}, UI\right) = G\left(\overline{es}_{p,m}\right) \cdot H_{|UI|+1}(UI) \cdot P\left(UI_{|UI|}, UI_{|UI|-1}, UI_{|UI|-2}, \ldots, UI_1\right),$$

so that for all possible combinations of UI,

$$P\left(R_{+|UI|+1}\right)=\sum\left[G\left(\overline{es}_{p,m}\right)\cdot H_{|UI|+1}\left(UI\right)\right]$$
$$\cdot P\left(UI_{|UI|},UI_{|UI|-1},UI_{|UI|-2},\ldots,UI_1,\varnothing\right)$$

$$using\left[G\left(\overline{es}_{p,m}\right)\cdot H_{\left|UI-\left\{R_{|UI|}\right\}\right|+1}\left(UI-\left\{R_{|UI|}\right\}\right)\right]$$

$$for\ P\left(R_{|UI|}\mid UI_{|UI|-1},UI_{|UI|-2},\ldots,UI_1,\varnothing\right),\ and$$

$$\left[1-G\left(\overline{es}_{p,m}\right)\cdot H_{\left|UI-\left\{NR_{|UI|}\right\}\right|+1}\left(UI-\left\{NR_{|UI|}\right\}\right)\right]$$

$$for\ P\left(NR_{|UI|}\mid UI_{|UI|-1},UI_{|UI|-2},\ldots,UI_1,\varnothing\right)$$

and so on until all the elements of *UI* have been addressed.

To provide an example of this process, consider the situation where two queue members stand between the intruder(s) and member of interest at queue position (+3). Let the number of members behind the (+3) position be given as *m*. Given $UI=\left\{UI_{+2},UI_{+1},\varnothing\right\}$, the probability of the member of interest reacting, $P\left(R_{+3}\right)$, given all possible combinations of *UI* is found by calculating,

$$\begin{aligned}P\left(R_{+3}\right)&=\sum\left[G\left(\overline{es}_{p,m}\right)\cdot H_{|UI|+1}\left(UI\right)\right]\cdot P\left(UI\right)\\&=G\left(\overline{es}_{p,m}\right)\cdot H_{+3}\left(R_{+2},NR_{+1},\varnothing\right)\cdot\left[G\left(\overline{es}_{p,m+1}\right)\cdot H_{+2}\left(NR_{+1},\varnothing\right)\right]\\&\cdot\left[1-G\left(\overline{es}_{p,m+2}\right)\cdot H_{+1}\left(\varnothing\right)\right]+G\left(\overline{es}_{p,m}\right)H_{+3}\left(R_{+2},R_{+1},\varnothing\right)\\&\cdot\left[G\left(\overline{es}_{p,m+1}\right)\cdot H_{+2}\left(R_{+1},\varnothing\right)\right]\cdot\left[G\left(\overline{es}_{p,m+2}\right)\cdot H_{+1}\left(\varnothing\right)\right]+G\left(\overline{es}_{p,m}\right)\\&\cdot H_{+3}\left(NR_{+2},NR_{+1},\varnothing\right)\cdot\left[1-G\left(\overline{es}_{p,m+1}\right)\cdot H_{+2}\left(NR_{+1},\varnothing\right)\right]\\&\cdot\left[1-G\left(\overline{es}_{p,m+2}\right)\cdot H_{+1}\left(\varnothing\right)\right]+G\left(\overline{es}_{p,m}\right)\cdot H_{+3}\left(NR_{+2},R_{+1},\varnothing\right)\\&\cdot\left[1-G\left(\overline{es}_{p,m+1}\right)\cdot H_{+2}\left(R_{+1},\varnothing\right)\right]\cdot\left[G\left(\overline{es}_{p,m+2}\right)\cdot H_{+1}\left(\varnothing\right)\right].\end{aligned}$$

The full process may be demonstrated in the relatively straightforward calculation of Condition 6: (+4) probability of reaction. Noting that $\overline{es}_H$ occurs with probability of 1 in Condition 6, and that

$$\overline{es}_{H,p=2,m=0}=0.780\cdot ln\left(e\cdot\frac{0+1.147}{1.147}\right)=0.780$$

$$\overline{es}_{H,p=2,m=1} = 0.780 \cdot ln\left(e \cdot \frac{1+1.147}{1.147} \right) = 1.269$$

$$\overline{es}_{H,p=2,m=2} = 0.780 \cdot ln\left(e \cdot \frac{2+1.147}{1.147} \right) = 1.567.$$

Using Table 3.8 with $Q\left(2\middle|3\middle|1\right) = \frac{10}{17}$, resulting in $p = 2$, $m = 1$, and noting for Condition 6 that $P(NR_{+2}, NR_{+1}) = 1$,

$$\begin{aligned} &P_{Q(2|3|1)}\left(R_{+4} \mid NR_{+3}, NR_{+2}, NR_{+1}\right) \\ &\quad = G\left(\overline{es}_{p=2,m=1}\right) \cdot H_{+4}\left(\left\{NR_{+3}, NR_{+2}, NR_{+1}\right\}\right) \\ &\qquad \cdot \left[1 - G\left(\overline{es}_{p=2,m=2}\right) \cdot H_{+3}\left(\left\{NR_{+2}, NR_{+1}\right\}\right)\right] \\ &\qquad \cdot P\left(NR_{+2}, NR_{+1}\right) = \left(1 - e^{-1.269}\right) \cdot e^{-\left[1.915 \cdot ln\left(\frac{3+1.553}{1.553}\right)\right]} \\ &\qquad \cdot \left[1 - \left(1 - e^{-1.567}\right) \cdot e^{-1.915 \cdot ln\left(\frac{2+1.553}{1.553}\right)}\right] = 0.076. \end{aligned}$$

$$\begin{aligned} &P_{Q(2|3|1)}\left(R_{+4} \mid R_{+3}, NR_{+2}, NR_{+1}\right) \\ &\quad = G\left(\overline{es}_{p=2,m=1}\right) \cdot H_{+4}\left(\left\{R_{+3}, NR_{+2}, NR_{+1}\right\}\right) \\ &\qquad \cdot G\left(\overline{es}_{p=2,m=2}\right) \cdot H_{+3}\left(\left\{NR_{+2}, NR_{+1}\right\}\right) \cdot P\left(NR_{+2}, NR_{+1}\right) \\ &\quad = \left(1 - e^{-1.269}\right) \cdot e^{-\left[1.298 \cdot ln\left(\frac{3+1.553}{2+1.553}\right) + 1.915 \cdot ln\left(\frac{2+1.553}{1.553}\right)\right]} \\ &\qquad \cdot \left(1 - e^{-1.567}\right) \cdot e^{-1.915 \cdot ln\left(\frac{2+1.553}{1.553}\right)} = 0.017 \end{aligned}$$

Similarly, from Table 3.8, we have $Q\left(2\middle|3\middle|0\right) = \frac{7}{17}$, so

$$
\begin{aligned}
&P_{Q(2|3|0)}\left(R_{+4} \mid NR_{+3}, NR_{+2}, NR_{+1}\right) \\
&\quad = G\left(\overline{es}_{p=2,m=0}\right) \cdot H_{+4}\left(\left\{NR_{+3}, NR_{+2}, NR_{+1}\right\}\right) \\
&\quad \cdot \left[1 - G\left(\overline{es}_{p=2,m=1}\right) \cdot H_{+3}\left(\left\{NR_{+2}, NR_{+1}\right\}\right)\right] \\
&\quad \cdot P\left(NR_{+2}, NR_{+1}\right) = \left(1 - e^{-0.780}\right) \cdot e^{-\left[1.915 \cdot ln\left(\frac{3+1.553}{1.553}\right)\right]} \\
&\quad \cdot \left[1 - \left(1 - e^{-1.269}\right) \cdot e^{-1.915 \cdot ln\left(\frac{2+1.553}{1.553}\right)}\right] = 0.059.
\end{aligned}
$$

$$
\begin{aligned}
&P_{Q(2|3|0)}\left(R_{+4} \mid R_{+3}, NR_{+2}, NR_{+1}\right) \\
&\quad = G\left(\overline{es}_{p=2,m=0}\right) \cdot H_{+4}\left(\left\{R_{+3}, NR_{+2}, NR_{+1}\right\}\right) \cdot G\left(\overline{es}_{p=2,m=1}\right) \\
&\quad \cdot H_{+3}\left(\left\{NR_{+2}, NR_{+1}\right\}\right) \cdot P\left(NR_{+2}, NR_{+1}\right) = \left(1 - e^{-0.780}\right) \\
&\quad \cdot e^{-\left[1.298 \cdot ln\left(\frac{3+1.553}{2+1.553}\right) + 1.915 \cdot ln\left(\frac{2+1.553}{1.553}\right)\right]} \cdot \left(1 - e^{-1.269}\right) \\
&\quad \cdot e^{-1.915 \cdot ln\left(\frac{2+1.553}{1.553}\right)} \\
&\quad = 0.012.
\end{aligned}
$$

$$
\begin{aligned}
\textit{Condition } 6\text{: } (+4)\, P\left(R_{+4}\right) &= \frac{10}{17} \cdot P_{Q(2|3|1)}\left(R_{+4}\right) + \frac{7}{17} \cdot P_{Q(2|3|0)}\left(R_{+4}\right) \\
&= \frac{10}{17} \cdot 0.093 + \frac{7}{17} \cdot 0.071 = 0.084.
\end{aligned}
$$

Besides becoming increasingly cumbersome, this process is likely to degrade in accuracy further from the intrusion point due to factors such as decreased likelihood of observing the intrusion, the observing member of interest missing the reaction of some of the intermediate members, or possibly the experimenter mistaking some reactions which seem possible for Condition 4: (+1) which was busiest from a queue member reaction standpoint. This approach does work quite well for the data evaluated, particularly for empirical reaction probabilities greater than 0.1, but further validation is necessary with possible improvements as additional data sets become available. Using this and previous mathematical modeling tools already presented, we can now solve for $\widehat{N}_0$ and $\overline{es}(\widehat{N}_0)$ using already derived Condition 1: (+1), 2: (+2), 5: (+2), and 6: (+3) sensation magnitude results and $H(UI)$ from Equation 3.5.

3.3.5 Estimating $\overline{es}_{H,1,0}$ and $\overline{es}_{H,2,0}$

Using Condition 1: (+1) and Condition 2: (+2) for $\overline{es}_{H,1,0}$ values and Condition 5: (+2) and Condition 6: (+3) for $\overline{es}_{H,2,0}$ values based on $H(UI)$, we have what is needed to derive values for $\widehat{N}_0$ and $\overline{es}(\widehat{N}_0)$ using Theorems 2.4 and 2.5. Conditions 1 and 2 involve a one-unit stimulus, and Conditions 5 and 6 involve two-unit stimuli. They are combined respectively to form the set $P = \{1,2\}$. In other words,

$$\overline{es}_{H,1} = \frac{22 \cdot 0.455 + 24 \cdot 0.455}{46} = 0.455 \; and \; \overline{es}_{H,2}$$
$$= \frac{20 \cdot 0.78 + 20 \cdot 0.78}{40} = 0.78.$$

Since the cumulative sample size of Conditions 1 and 2 is comparable with Conditions 5 and 6 (i.e., 46 versus 40 respectively), we can better justify applying Theorem 2.5 to find $\widehat{N}_0$ such that

$$\left[0.455 - ln\left(\frac{1+\widehat{N}_0}{\widehat{N}_0}\right) \cdot \frac{1}{1} \cdot \frac{0.780}{ln\left(\frac{2+\widehat{N}_0}{\widehat{N}_0}\right)}\right]$$
$$+\left[0.780 - ln\left(\frac{2+\widehat{N}_0}{\widehat{N}_0}\right) \cdot \frac{1}{1} \cdot \frac{0.455}{ln\left(\frac{1+\widehat{N}_0}{\widehat{N}_0}\right)}\right] = 0$$

TABLE 3.10 Summary of Empirical Versus Theoretical Results for Schmitt et al. (1992) and Milgram et al. (1986) Field Experiments with Permission from the American Psychological Association

Condition	*Position in Queue*	*# Obs.*	*Field Experiment Reaction Probability*	*95% Credibility Interval*	*Theoretical Reaction Probability*
Illegitimate High	(+1)	30	0.600	(0.422, 0.755)	0.599
Illegitimate Low	(+1)	30	0.366	(0.054, 0.298)	0.366
Condition 1	(+1)	22	0.364	(0.197, 0.572)	0.364
1	(+2)	14	0.143	(0.043, 0.405)	0.170
1	(+3)	9	0.0	(0.000, 0.259)	0.100
Condition 2	(+2)	24	0.167	(0.068, 0.361)	0.167
2	(+3)	15	0.0	(0.000, 0.171)	0.104
Condition 3	(+3)	20	0.0	(0.000, 0.133)	0.063
Condition 4	(+1)	23	0.870	(0.676, 0.952)	0.817
4	(+2)	23	0.435	(0.255, 0.633)	0.401
4	(+3)	22	0.091	(0.028, 0.280)	0.252
Condition 5	(+2)	20	0.200	(0.082, 0.419)	0.200
5	(+3)	15	0.0	(0.000, 0.171)	0.094
Condition 6	(+3)	20	0.150	(0.054, 0.363)	0.150
6	(+4)	17	0.118	(0.036, 0.347)	0.084

Solving this for social noise in a New York City queue results in $\widehat{N}_0 = 2.056$. The value for $\overline{es}\left(2.056\right)$ is then,

$$\overline{es}\left(2.056\right) = \frac{1}{2} \cdot \left[\frac{0.455}{ln\left(\frac{1+2.056}{2.056}\right)} + \frac{0.780}{ln\left(\frac{2+2.056}{2.056}\right)} \right] = 1.148.$$

TABLE 3.11 Summary Results of Unit Sensation Magnitude Values by Situation

Social Situation	$\bar{es}$
Illegitimate Intrusion – Low-Unit Stimulus Sensation Magnitude	$\bar{es}_L = 0.2869$
Crowd-Gathering (CG) Experiment – New York City Sidewalk	$\bar{es}_{CG} = 0.5890$
Illegitimate Intrusion – Grand Central Rail Ticket Counter	$\bar{es}_H = 1.1477$

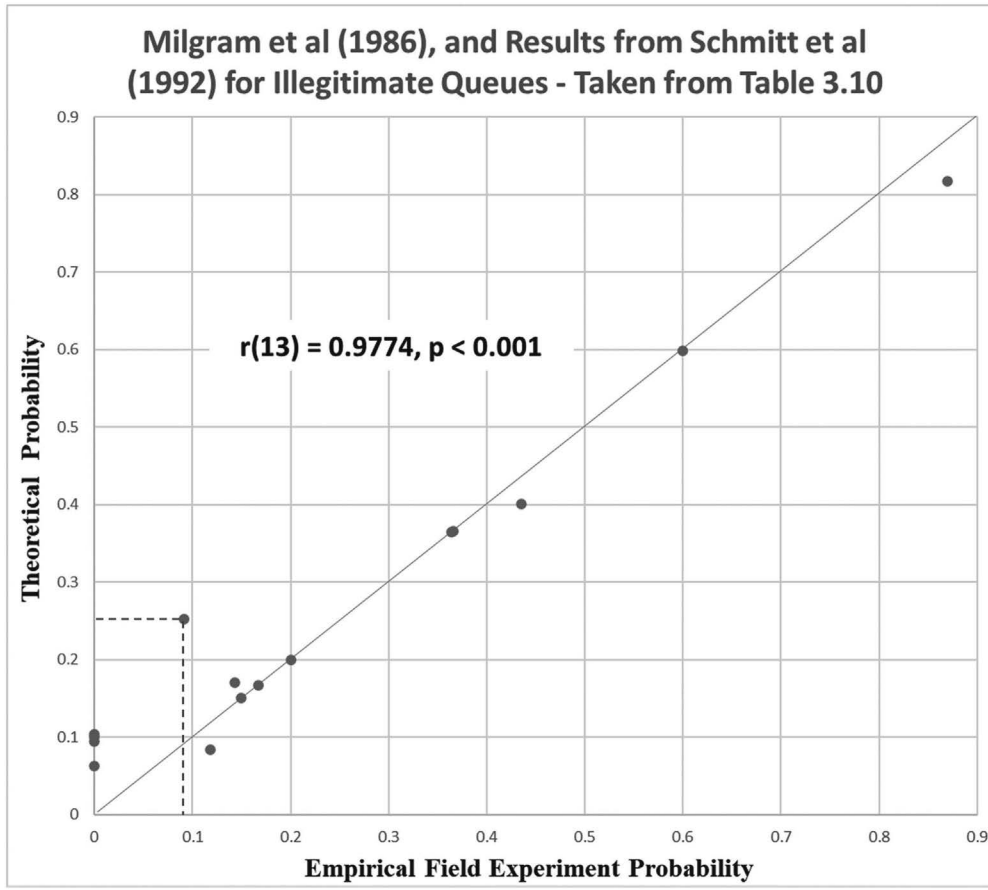


FIGURE 3.1 Empirical Results from Milgram et al. (1986) and Schmitt et al. (1992) versus Theoretically Derived Results for the Same.

Therefore, it should be no surprise using $\widehat{N}_0 = 2.056$ that

$$\overline{es}_{H,1} = 1.148 \cdot ln\left(\frac{1+\widehat{N}_0}{\widehat{N}_0}\right) = 0.455 \text{ and}$$

$$\overline{es}_{H,2} = 1.148 \cdot ln\left(\frac{2+\widehat{N}_0}{\widehat{N}_0}\right) = 0.780, \text{ for } \widehat{N}_0 = 2.056.$$

In probability space, when applying these results, $\widehat{N}_0 = 2.056$, and the derived values for $H(UI)$ to Conditions 1: (+1), 2: (+2), 5: (+2), and 6: (+3) a mean square error of 0.0000018 is obtained. This is anticipated but stated to emphasize when viewing Table 3.10 that social queue model parameters were trained on these four conditions and positions. Additional conditions and positions from Milgram et al. (1986) having field experiment reaction probabilities greater than zero could have been used, but these four met Corollary 2.3 requirements and have used the fewest assumptions in their theoretical calculation. Hence, it is felt that their use represents the best that can be obtained with the empirical data available.

The values have been established for $\overline{es}_H$, $\overline{es}_L$. $\widehat{N}_0$, $\widehat{N}_{UA}$, $\widehat{N}_{UI}$, with the functions $G_{ES}(es_{p,m})$ and $H(UI)$ defined for use in creating the theoretical reaction probabilities contained in Table 3.10. All trial configurations fall within their 95 percent credibility intervals. The variable values as shown here are used to calculate the Table 3.10 theoretical reaction probability calculations.

$$\overline{es}_H = 1.148 \qquad \overline{es}_L = 0.287;$$

$$\widehat{N}_0 = 2.056, \qquad \widehat{N}_{UA} = 1.147 \qquad \widehat{N}_{UI} = 1.553 \text{ (}\textit{possibly constant}\text{)};$$

$$G_{ES}\left(es_{p,m}\right) = 1 - e^{-es_p \cdot ln\left(e \cdot \frac{m+N_{UA}}{N_{UA}}\right)}$$

$$\text{and } H_{|UI|+1}\left(UI\right) = e^{-\sum_{i=0}^{|UI|-1} k\left(UI_{i+1}\right) \cdot ln\left(\frac{|UI|-i+noise}{|UI|-(i+1)+noise}\right)}$$

where $k(R_{i+1}) = 1.298$ a*nd* $k(NR_{i+1}) = 1.915$.

Table 3.11 provides a summary of unit stimulus sensation magnitude values derived so far as a function of the social situation. We have already shown that $\overline{es}_H \cong 4 \cdot \overline{es}_L$. The unit sensation magnitude from the crowd-gathering experiment indicates an integer relation as well, namely $\overline{es}_H \cong 2 \cdot \overline{es}_{CG} \cong 4 \cdot \overline{es}_L$. Whether odd coincidence, or an indication of the existence of JNDs, more data in varying situations would be required to gain greater clarity. Until then, Table 3.11 highlights a possible pattern, providing yet another potential direction for investigation.

Empirical results from the field experiments conducted by Milgram et al. (1986) and Schmitt et al. (1992) are compared to theoretical results as developed in this

chapter. Figure 3.1 provides a visual depiction of how the two sets of results compare, keeping in mind that a good portion of the empirical reaction probabilities greater than 0.1 were used in determining model variable values. This indicates nothing more than a very good start, with necessary improvements anticipated. Finally, as a marker supporting future effort and comparison, a linear regression performed on the two sets indicates a very strong direct relationship between empirical and theoretical results overall. Note the Pearson's $r(13)$ value of 0.9774 with $p < 0.001$.

With the tools in place and the very strong correlation of calculated results versus empirical results from the two independent queue intrusion experiments, we are now ready to venture outside the queue to determine if the model as derived has any operational utility for social systems that may be transformed into a queue-like hierarchical process. Before doing so, it would be useful to summarize the findings and provide some thoughts on the results before proceeding.

3.4 Summary

Mathematical tools, developed in Chapters 1, 2, and 3, are to be carried forward for evaluation against more socially diverse situations that go beyond the queue as a basic social system. As indicated at the beginning of this book, modeling the queue as a social system provides a means to understand a particular social subgroup based on its social norms. With this understanding, we may be able to better interpret more complex social interactions within the culture of interest, in the present case the western culture with its own social norms, and build on what is developed here. Chapter 4 will begin to consider social scenarios outside of the queue, scenarios which may be transformed into a queue-like process for analysis.

3.4.1 Observations

There are three observations to be made before continuing. The first is the potential progression of development and structure of the derived model as presented. The second involves additional thoughts regarding the Golden Ratio and its appearance in the Milgram et al. (1986) data. The third and final observation is how modeling of the social queue may also apply to more complex social situations, such as may be found in politics and political identification. These are all observations highlighted for future consideration as more is learned about human brain development and operation, and more experimental data having an underlying queue structure becomes available.

As the first observation, the model as derived may represent three distinct stages of complexity. The first, under uniform encoding, accommodates a finite number of social stimuli and accounts for group pressure to support enforcement of social norms. Uniform encoding may have been all that was necessary in prehistoric times with relatively small bands of hunter-gatherers (Dunbar and Sosis, 2018). As human populations grew and became concentrated in city states, the transition from uniform encoding to exponential encoding may have become necessary as a result of social evolution. The transition allows for the existence of infinitely many stimuli and just as many group members to amplify the sensation magnitude caused by the stimuli. The third and final stage is represented by

the development of an independent means to gain uncertainty-based information through member reactions that reduce uncertainty. When combined, it seems human evolution may have shifted or may be shifting focus from individual organism survival to group organism survival enhancements. In other words, the group functioning as a larger organism has a better chance of survival than that of the individual. Maybe this is reading too much into what has been observed during model development, but it does seem to warrant further consideration, given the work on multilevel selection theory by Wilson (2015, pp. 47–48) and Cheng (2020).

Equation 3.5 brings us back to Stanley Stevens' power law and the second observation. What seems logical in exponential space may seem more interesting in power space. In this case, recalling the Golden Ratio equals 1.618, and that its inverse is 0.618, consider that

$$\left(\frac{1.618}{1+1.618}\right)^4 = (0.618)^4 = 0.146 \cong e^{-z(3)} = 0.127;$$

$$\left(\frac{1.618}{1+1.618}\right)^3 = (0.618)^3 = 0.236 \cong e^{-z(2)} = 0.205;$$

$$\left(\frac{1.618}{1+1.618}\right)^2 = (0.618)^2 = 0.382 \cong e^{-z(1)} = 0.386$$

These observations remind us of Benjafield and Adams-Weber (1976) and Gross and Miller (1997) and their hypothesis that the Golden Section plays a role in human processing of information. Applying this to Arons and Irwin (1932), in their experiment, they had subjects indicate if a second subsequent weight was lighter, the same, or heavier than a standard weight provided for comparison. In all cases, without the subjects knowing, the second weight was the same as the standard weight, based on 2,500 trials, subject H indicated with a 0.2628 probability the second weight was lighter, with a 0.3156 probability the second weight weighed the same, and with a 0.4216 probability the second weight was heavier. Interestingly enough

$$\frac{0.4216}{0.2628} = 1.604 \cong 1.618.$$

For the third observation, if the postulate is correct that observation of group member reactions is performed by the member of interest to reduce uncertainty regarding the severity of the event, and given that most queues in Milgram et al. (1986) having a theoretical probability of reaction of about 0.1 or less had no reactions directed at the intruder during the experiment, then if four group members in sequence immediately behind the intrusion point do not react, no one remaining behind those four are likely to react. This is shown by,

$$\left(\frac{1.618}{1+1.618}\right)^2 \cdot \left(\frac{1+1.618}{2+1.618}\right)^2 \cdot \left(\frac{2+1.618}{3+1.618}\right)^2 \cdot \left(\frac{3+1.618}{4+1.618}\right)^2$$
$$= \left(\frac{1.618}{4+1.618}\right)^2 \cong e^{-z(4)} = 0.0874$$

This should certainly be true after the fifth member resulting in $e^{-z(5)} = 0.064$. This supports the localized nature of reactions as commented on in Milgram et al. (1986, pp. 688–689) and indicates that multiple members become involved, implying that it is not up to just one or two members to inform the rest of the group. What this also means is that if three of four levels of hierarchy responsible for informing the remainder of the group do not react intentionally for whatever purpose, they can for the most part control uncertainty-based information used by the other in-group members they are responsible to. Control of information, by reaction, non-reaction, or other means, relates to control of perception and how a cohesive group interprets an event. If the larger group is not cohesive, then consider the more cohesive and smaller subgroups whose members will depend on the social attractiveness of the leaders they choose within a hierarchical structure for interpretation of the event.

If social leaders can form and control a highly attractive (i.e., cohesive) group of members who share similar views, then the group and its beliefs can become self-perpetuating as alluded to by Festinger (1957, pp. 186–187, with permission from Stanford University Press),

> *What is more, the degree to which the persistent deviant was rejected was greater in the high attraction than in the low attraction groups. In other words, there was evidence that reduction of dissonance was attempted through rejection of the person who voiced disagreement and that the extent to which this occurred depended upon the magnitude of the dissonance created by this disagreement.*

This chapter finishes the queue model derivation. To investigate possible applications of this model in more complex social situations, we now move outside of the queue and consider social systems that may be transformed back into a queue-like social structure, some of which have already been alluded to. To be a little mischievous, this transformation approach will be termed the queue transform, given it transforms, in appropriate cases, one social system into another. This is similar in nature to the Constant-Q transform used in signal processing used to transform data series information into the frequency domain where the information is more meaningfully and easily evaluated.

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